QUESTION BANK FOR WEEK9 TO WEEK14 TOPICS

QUESTION BANK ON SOCKET PROGRAMMING & PROCESS

PROCESS: BASIC DATA STRUCTURE & SYSTEM CALLS (CHAPTER 3)

1) What is a process?
2) What are Lightweight Processes in Linux?
3) Explain what is meant by multithreaded application. Describe how multithreaded applications are implemented in linux. Give three examples of POSIX-compliant pthread libraries.
4) Describe thread group in linux.
5) What is meant by process descriptor and explain the role of the process descriptor?
6) With the help of a schematic, describe the Linux process descriptor struct task_struct.
7) List the seven possible process states in Linux OS and explain.
8) Describe the following process states in linux :
TASK_RUNNING
TASK_INTERRUPTIBLE
TASK_UNINTERRUPTIBLE
TASK_STOPPED
TASK_TRACED
EXIT_ZOMBIE
EXIT_DEAD
9) State the purpose of TASK_UNINTERRUPTIBLE state in linux process.
10) State the two approaches used by kernel to identify a process.
11) What is meant by PID? Why is it required?
12) What is the upper limit on the PID values in 32-bit architectures?
13) What is the upper limit on the PID values in 64-bit architectures?
14) State the purpose of a pidmap_array bitmap in linux. Indicate the size of the pidmap_array bitmap.
15) Indicate which lightweight process becomes by default the thread group leader. What is the PID assigned to the thread group?
16) Since kernel must be able to handle many processes at the same time, hence process descriptors are stored in .... memory rather than in the memory area permanently assigned to the kernel.
17) Describe the two different data structures assigned to every process by the kernel in a single per-process memory area. Indicate the length of this memory area and explain the data structures stored in this memory area with the help of a diagram showing clearing both these data structures.

18) Explain the purpose of the following two different data structures assigned to every process by the kernel in a single per-process memory area. Indicate the size of each of these data structures.
a) Thread_info structure b) kernel mode process stack
19) State the key benefit in terms of efficiency offered by the kernel in providing a close association between the thread_info structure and the kernel mode stack.
20) State the purpose of current macro in linux.
21) Explain what is meant by process list. Explain how process list is implemented in linux.
22) With the help of a block diagram, show the special data structures used to implement the process list. Also indicate the processes the first two nodes in the data structure point to.
23) Explain how runqueue is implemented in linux to achieve scheduler speedup to select the best runnable process in constant time.
24) With the help of a figure, illustrate the relationship between parent and siblings of a group of processes. Process P0 successively created P1, P2, and P3. Process P3, in turn, created process P4
25) Write short notes on pid hash table and chained lists.
26) With the help of block diagram, describe pid hash table and chained lists.
27) What are the four PID hash tables used in linux. Why are they required?
28) With the help of a block diagram, describe the four hash tables and show how these data structures are implemented showing clearly the chained lists and processes in each group.
29) Explain the purpose of wait queue in linux.
30) Describe runqueue and wait queues in Linux.
31) Indicate the three important uses of the wait queues in the kernel.
32) Describe how wait queues are implemented in linux by providing the following two data structures - struct __wait_queue_head and struct wait_queue and clearly indicating the purpose of various fields in these data structures.
33) What is meant by "thundering herd" problem. Explain how it is tackled.
34) What are the two kinds of sleeping processes in wait queue. Why do you require this classification.
35) Explain process Resource limits in Linux and list the various resource limits specified.
36) State the purpose of the following commands in unix/linux
a) ulimit -Ha b) ulimit -Sa c) getconf -a d) getconf PAGE_SIZE
37) Describe Process switch or context switch or task switch in linux.
38) Explain Hardware context in Linux.
39) Write short notes on the following :
a) Task State Segment b) Thread field
40) Describe how kernel performs a process switch using schedule() function.
41) Explain three different mechanisms introduced by modern Unix kernels to solve the problem of duplicating address space. Also indicate the system call associated with each one of these mechanisms to create a process.
42) State the three system calls used to create a process. Why are these system calls required?

43) Write the system call used to create a lightweight process in linux.
44) Write short notes on
a) Process 0 b) Process 1
45) Describe the operations performed by the following functions in linux :
a) do_fork() b) copy_process()
46) What are kernel threads? Explain how kernel threads are created in linux.
47) State the purpose of following functions in linux.
a) kernel_thread() b) copy_process() c) exit() d) exit_group()
48) Describe how processes are created and terminated/destroyed in Linux.
49) What is orphan process? How orphan process is handled in linux to eliminate memory leak?
50) What happens if parent process terminate before its children? How is this problem solved to eliminate memory leaks?

SOCKET PROGRAMMING

1. Explain what is meant by a socket.
2. Describe socket programming.
3. With the help of a neat schematic, describe the client and server model.
4. List out the steps involved in establishing a socket on the server side.
5. List out the steps involved in establishing a socket on the client side.
6. Discuss TCP/IP Protocol suite
7. Write the specific functions performed by TCP and IP protocols in the internet protocol suite.
8. What is an IP address? Distinguish between IPv4 and IPv6 addresses.
9. Explain the following:
a) Address Domain b) Address format c) Port Number d) Socket types
10. What are well-known ports or default TCP port numbers?
11. Specify the default Port numbers associated with the following :
Telnet, FTP Data, FTP control, SSH, SMTP, DNS, HTTP, SNMP, HTTPS DHCP server, DHCP client
12. Describe the following functions by providing the complete syntax of the function and parameters required :
a) Socketb) setsockopt c) bind d) listen e) accept
f)connect g) read h) write i) send j) recv
13. State the purpose of select() command. Provide the complete syntax of the command clearly specifying the parameters required.

QUIZ QUESTIONS :

Question 1
The thread_info structure is .... bytes long, the kernel stack can expand up to .... bytes.
Select one:
1. 192, 8000
2. 52, 8140
3. 64, 8128
4. 92, 8100
Question 2
The kernel thread created by process 0 executes the .... function, which in turn completes the initialization of the kernel.
Select one:
1. copy_process()
2. init()
3. do_fork()
4. schedule()
Question 3
The child process rarely needs to read or modify all the resources inherited from the parent; in many cases, it issues an immediate .... and wipes out the address space that was so carefully copied.
Select one:
1. execve
2. fork
3. vfork
4. clone

Question 4
Version 4 of the Internet Protocol (IPv4) defines an IP address as a .... number.
Select one:
1. 64-bit
2. 32-bit
3. 128-bit
4. 256-bit
Question 5
Identify the layers of the OSI Model ....
Select one or more:
1. Application Layer
2. Network Layer
3. Physical Layer
4. Presentation Layer
5. Data Link Layer
6. Session Layer
7. Transport Layer
8. Network Interface Layer
9. Internet Layer

Question 6
The .... flag in the CR0 allows the kernel to save and restore the FPU, MMX, and XMM registers only when really needed.
Select one:
1. PG
2. TF
3. PS
4. TS
Question 7
A process waiting for a resource that can be granted to just one process at a time is a typical .... process.
For each process, Linux packs two different data structures in a single per-process memory area: a small data structure linked to the process descriptor.
What are they...........
Select one or more:
1. Task State Segment
2. Kernel Mode process stack
3. thread_info structure
4. User Mode process stack
Question 9
When a User Mode process attempts to access an I/O port by means of an in or out instruction, the CPU may need to access an I/O Permission Bitmap stored in the .... to verify whether the process is allowed to address the port.
Question 10
.... field in the process descriptor of a process P Points to the head of the list containing all children created by P.

Question 11
The following Resource limit field in process descriptor indicates maximum size of process address space in bytes
Select one:
1. RLIMIT_DATA
2. RLIMIT_RSS
3. RLIMIT_AS
4. RLIMIT_DATA

Question 12
Specify the default Port number associated with Telnet ....
Question 13
The .... function operates on the current process: The function sets the state of the current process to TASK_UNINTERRUPTIBLE and inserts it into the specified wait queue.
Select one:
1. interruptible_sleep_on( )
2. sleep_on_timeout( )
3. prepare_to_wait_exclusive( )
4. sleep_on( )

Question 14
In multiprocessor systems, ... process running on CPU 0 initializes the kernel data structures, then enables the other CPUs and creates the additional .... processes by means of the copy_process( ) function passing to it the value 0 as the new PID. You need to specify only one answer which satisfies both the fill in the blanks
Question 15
The kernel must be able to handle many processes at the same time, and process descriptors are stored in ....
Select one:
1. flash memory
2. kernel memory
3. dynamic memory
4. read only memory
Question 16
.... defines how applications can create channels of communication across a network. It also manages how a message is assembled into smaller packets before they are then transmitted over the internet and reassembled in the right order at the destination address.
Select one:
1. IP
2. ARP
3. TCP
4. HTTP
Question 17
.... sleeping processes are selectively woken up by the kernel when the event occurs.
Select one:
1. Non-Exculsive
2. Ready
3. Exclusive
4. Runnable
Question 18
.... field in the process descriptor of a process P points to the process descriptor of the process that created P or to the descriptor of process 1 (init) if the parent process no longer exists.
Question 19
Identify the Layers in TCP/IP Model ....
Select one or more:
1. Physical Layer
2. Data Link Layer
3. Application Layer
4. Presentation Layer
5. Transport Layer
6. Network Interface Layer
7. Network Layer
8. Internet Layer
9. Session Layer

Question 20
…………………… command is used in Linux to list the resource limits.
Select one:
1. top
2. df
3. ulimit -a
4. du

Question 21
In Linux PROCESS LIST is implemented as …….
Select one:
1. singly linked list
2. binary search tree
3. Merkel Patricia tree
4. doubly linked list

Question 22
When an 80x86 CPU switches from User Mode to Kernel Mode, it fetches the address of the Kernel Mode stack from the ………..

Question 23
Each element in the wait queue list represents a sleeping process, which is waiting for some event to occur; its descriptor address is stored in the ………. field of wait queue element structure.

Question 24
If any parent process terminates without reaping a child, then the orphaned child will be reaped by ……………………
Select one:
1. Zombie process
2. init process
3. swapper process
4. process scheduler

Question 25
Control flow passes from one process to another process via ……………………….
Select one:
1. Return instruction
2. Context switching
3. Jump instruction
4. Call instruction

Question 26
In Linux, depending on the process priority, the runqueue list has been split into ……………………different lists in order to make process scheduler operations more efficient.
Select one:
1. 70
2. 3
3. 140
4. 6

Question 27
.... is the final state which indicates that the process is being removed by the system because the parent process has just issued a wait4( ) or waitpid( ) system call for it.
Select one:
1. TASK_RUNNING
2. EXIT_ZOMBIE
3. TASK_TRACED
4. EXIT_DEAD
5. TASK_INTERRUPTIBLE
6. TASK_UNINTERRUPTIBLE
7. TASK_STOPPED

Question 28
The upper limit on the PID number in 32-bit architectures is ....
Select one:
1. 32767
2. 32768
3. 2048
4. 1024

Question 29
Linux Kernel uses .... function to perform process switch.
Select one:
1. kernel_thread()
2. do_fork()
3. vfork()
4. schedule()

Question 30
The size of the pidmap_array bitmap used in 32-bit architectures to indicate which PIDs are currently assigned and free is .... bits.
Select one:
1. 32767
2. 4096
3. 65536
4. 32768

Question 31
Exclusive processes are identified by the value ..... in the .... field of the corresponding wait queue element.
Select one:
1. 1, func
2. 0, flags
3. 1, flags
4. 0, func

Question 32
Specify the default Port number associated with Finger .....
Question 33
Maximum number of pages used for storing pidmap_array bitmap in 32-bit architectures to indicate which PIDs are currently assigned and free is.......
Select one:
1. 1
2. 8
3. 128
4. 4

Question 34
Process is ....
Select one:
1. nothing but contents of main memory
2. a job in secondary memory
3. an instance of a program in execution
4. a program in High level language kept on disk

Question 35
If a child process terminates before the parent process dies then the child process is termed as ....
Select one:
1. Bad process
2. Zombie process
3. Orphan process
4. Dead process

Question 36
In Linux a .... is basically a set of lightweight processes that implement a multithreaded application and act as a whole with regards to some system calls such as getpid( ), kill( ), and _exit( )

Question 37
Maximum number of PIDs that can be used in in 64-bit architectures is ....
Select one:
1. 65536
2. 4194303
3. 4194304
4. 32768

Question 38
.... field in the process descriptor of a process P Points to the current parent of P (this is the process that must be signaled when the child process terminates); its value usually coincides with that of real_parent. It may occasionally differ, such as when another process issues a ptrace( ) system call requesting that it be allowed to monitor P

Question 39
Non-Exclusive processes are identified by the value ..... in the .... field of the corresponding wait queue element.
Select one:
1. 0, func
2. 1, flags
3. 1, func
4. 0, flags

Question 40
.... defines how to address and route each packet to make sure it reaches the right destination.
Select one:
1. ARP
2. SNMP
3. TCP
4. IP

Question 41
.... system call allows both the parent and the child to share many per-process kernel data structures, such as the paging tables (and therefore the entire User Mode address space), the open file tables, and the signal dispositions.
Select one:
1. execve
2. clone
3. fork
4. vfork

Question 42
Specify the default Port number associated with DNS ....

Question 43
Linux associates a different PID with each process or lightweight process in the system. Say True or False
Select one:
True
False

Question 44
In Linux, the .... function is used to handle the clone( ), fork( ), and vfork( ) system calls
Select one:
1. kernal_thread
2. execve
3. copy_process
4. do_fork

Question 45
Stream sockets use .... protocol which is a reliable, stream oriented protocol (connection-oriented)
Select one:
1. UDP
2. IP
3. SNMP
4. TCP

Question 46
If a child process terminates after the parent process dies then the child process is termed as ....
Select one:
1. Zombie process
2. Orphan process
3. Bad process
4. Dead process

Question 47
.... is a kernel thread created from scratch during the initialization phase of Linux .
Select one:
1. Init process
2. schedule
3. kirq daemon
4. Swapper process
Question 48
The following Resource limit field in process descriptor indicates maximum number of page frames owned by the process.
Select one:
1. RLIMIT_MEMLOCK
2. RLIMIT_RSS
3. RLIMIT_AS
4. RLIMIT_DATA
Question 49
A .... represents a set of sleeping processes, which are woken up by the kernel when some condition becomes true.
Select one:
1. wait queue
2. ready queue
3. run queue
4. stack
Question 50
PID of a newly created process is normally the PID of the previously created process increased by one.
Say True or False

Question 51
The following Resource limit field in process descriptor indicates maximum CPU time for the process in seconds.
Select one:
1. RLIMIT_AS
2. RLIMIT_RSS
3. RLIMIT_CPU
4. RLIMIT_DATA

Question 52
...........................uses Copy On Write (COW) technique that allows both the parent and the child processes to read the same physical pages. Whenever either one tries to write on a physical page, the kernel copies its contents into a new physical page that is assigned to the writing process.
Select one:
1. vfork() system call
2. clone() system call
3. fork() system call
4. flock() system call

Question 53
Linux associates a different .... with each process or lightweight process in the system. This approach allows the maximum flexibility, because every execution context in the system can be uniquely identified.

Question 54
For systems having 512 MB of RAM, each hash table is stored in .... page frames and includes .... entries.
Select one:
1. 2, 2048
2. 8, 4096
3. 4, 4096
4. 4, 2048

Question 55
The system call used to create a light-weight process or thread is ....
Select one:
1. fork
2. flock
3. vfork
4. clone

Question 56
.... field of the process descriptor describes what is currently happening to the process.
Select one:
1. run_list
2. tasks
3. state
4. thread_info

Question 57
.... state is used to indicate that the process is suspended (sleeping) until some condition becomes true. Raising a hardware interrupt, releasing a system resource the process is waiting for, or delivering a signal are examples of conditions that might wake up the process
Select one:
1. EXIT_DEAD
2. EXIT_ZOMBIE
3. TASK_TRACED
4. TASK_STOPPED
5. TASK_RUNNING
6. TASK_UNINTERRUPTIBLE
7. TASK_INTERRUPTIBLE
Question 58
The .... register of each CPU contains the TSSD Selector of the corresponding TSS.
Select one:
1. tr
2. idtr
3. ldtr
4. gdtr
Question 59
To speed up the search for a process descriptor corresponding to a given PID, ... hash tables have been introduced in Linux.
Select one:
1. 4
2. 8
3. 2048
4. 16

Question 60
A .... is an end point of communication between two systems on a network.

Question 61
The maximum number of pages allocated for storing PID bitmap in 64-bit architectures is ....
Select one:
1. 128
2. 1
3. 8
4. 64
Question 62
The system call used to create a light-weight process or thread is ....
Select one:
1. vfork
2. flock
3. Clone
4. fork

Question 63
Indicate the correct order of the steps involved in establishing a socket on the server side:
1. Connect
2. Accept
3. Socket Creation
4. Listen
5. Send/Recv
6. Set Socket option
7. Bind
Select one:
1. 3, 6, 7, 4, 2, 5
2. 3, 6, 7, 2, 4, 5
3. 3, 7, 6, 4, 2, 5
4. 3, 6, 2, 7, 4, 5

Question 64
What is the full form of NGPT which is a package from IBM for implementing multithreading applications

Question 65
In Linux, a task_struct type structure is used to store ....
Select one:
1. segment descriptor
2. process descriptor
3. page descriptor
4. file descriptor

Question 66
The..................... function initializes all the data structures needed by the kernel, enables interrupts, and creates another kernel thread, named process 1 (init process)
Select one:
1. setup()
2. first startup()
3. second startup()
4. start_kernel( )

Question 67
The command getconf PAGE_SIZE returns ....
Select one:
1. 16384
2. 8192
3. 65536
4. 4096

Question 68
……….. process state is used under certain specific conditions in which a process must wait until a given event occurs without being interrupted.
Select one:
1. TASK_STOPPED
2. TASK_UNINTERRUPTIBLE
3. TASK_INTERRUPTIBLE
4. TASK_RUNNING
5. TASK_TRACED
6. EXIT_ZOMBIE
7. EXIT_DEAD

Question 69
The function to get the PID of a calling process is ……………………..
Select one:
1. getpid
2. getppid
3. getpgrp
4. getppgrp
Question 70
The ………. register of each CPU contains the TSSD Selector of the corresponding TSS.
Select one:
1. ldtr
2. gdtr
3. tr
4. idtr

Question 71
Specify the default Port number associated with POP3 …………

Question 72
The set of data that must be loaded into the registers before the process resumes its execution on the CPU is called the ……………………
Select one:
1. context switch
2. task switch
3. software context
4. hardware context

Question 73
All …………………… sleeping processes are always woken up by the kernel when the event occurs.
Select one:
1. Ready
2. Non-Exclusive
3. Exclusive
4. Runnable

Question 74
The .... function sets up the process descriptor and any other kernel data structure required for a child's execution.
Select one:
1. kernal_thread
2. execve
3. do_fork
4. copy_process

Question 75
Datagram sockets use .... protocol, which is unreliable and message oriented. (Connection less)
Select one:
1. TCP
2. ARP
3. UDP
4. IP

Question 76
Specify the default Port number associated with HTTP ....

Question 77
The following Resource limit field in process descriptor indicates maximum size of nonswappable memory in bytes.
Select one:
1. RLIMIT_RSS
2. RLIMIT_DATA
3. RLIMIT_AS
4. RLIMIT_MEMLOCK

Question 78
.... state indicates whether process is either executing on a CPU or waiting to be executed.
Select one:
1. EXIT_DEAD
2. TASK_RUNNING
3. EXIT_ZOMBIE
4. TASK_UNINTERRUPTIBLE
5. TASK_INTERRUPTIBLE
6. TASK_STOPPED
7. TASK_TRACED

Question 79
In Linux, a part of the hardware context of a process is stored in the .... field of the process descriptor while the remaining part is saved in the Kernel Mode stack.

Question 80
.... system call creates a process that shares the memory address space of its parent. To prevent the parent from overwriting data needed by the child, the parent's execution is blocked until the child exits or executes a new program.
Select one:
1. cfork
2. fork
3. clone
4. vfork

Question 81
The .... field of a wait queue element is used to specify how the processes sleeping in the wait queue should be woken up.
Select one:
1. func
2. task_list
3. flags
4. task

Question 82
NPTL stands for ....

Question 83
In 64-bit architectures, the system administrator can enlarge the maximum PID number up to ....
Select one:
1. 4,194,303
2. 32768
3. 4,194,304
4. 65536

Question 84
.... field in the process descriptor of a process P have pointers to the next and previous elements in the list of the processes, those that have the same parent as P.

Question 85
.... state indicates that Process execution has been stopped by a debugger and process is being monitored by debugger.
Select one:
1. EXIT_ZOMBIE
2. EXIT_DEAD
3. TASK_STOPPED
4. TASK_RUNNING
5. TASK_UNINTERRUPTIBLE
6. TASK_INTERRUPTIBLE
7. TASK_TRACED

Question 86
Specify the default Port number associated with SQL Server ....

Question 87
Specify the default Port number associated with HTTPS ....

Question 88
The 80x86 microprocessors automatically save the FPU, MMX, and XMM registers in the TSS.
SAY TRUE OR FALSE

Question 89
Wait queues are implemented as.............. whose elements include pointers to process descriptors.
Select one:
1. Merkel Patricia Tree
2. singly linked lists
3. doubly linked lists
4. binary search trees

Question 90
The state field of the process descriptor indicates one of the possible .... process states.
Select one:
1. 4
2. 5
3. 8
4. 7

Question 91
The aim of using large number of runqueus is to allow the scheduler to select the best runnable process in constant time, independently of the number of runnable processes.
Say True or False

Question 92
Specify the default Port number associated with FTP - Data

Question 93
The following Resource limit field in process descriptor indicates maximum number of open file descriptors
Select one:
1. RLIMIT_DATA
2. RLIMIT_MEMLOCK
3. RLIMIT_NOFILE
4. RLIMIT_NPROC

Question 94
The resource limits for the current process are stored in the current->signal->....... field, that is, in a field of the process's signal descriptor
Question 95
The following Resource limit field in process descriptor indicates the maximum heap size, in bytes
Select one:
1. RLIMIT_AS
2. RLIMIT_FSIZE
3. RLIMIT_DATA
4. RLIMIT_RSS

Question 96
.... is a problem wherein multiple processes are wakened only to race for a resource that can be accessed by only one of them, with the result that remaining processes must once more be put back to sleep.
Select one:
1. Race Condition
2. Livelock
3. Cache Coherency
4. Thundering herd
Question 97
The .... system call causes the process to block until a client connects to the server. Thus, it wakes up the process when a connection from a client has been successfully established. It returns a new file descriptor, and all communication on this connection should be done using the new file descriptor.

Question 98
The following Resource limit field in process descriptor indicates maximum number of processes that the user can own.
Select one:
1. RLIMIT_MEMLOCK
2. RLIMIT_NPROC
3. RLIMIT_DATA
4. RLIMIT_NOFILE

Question 99
Linux uses .... processes to offer better support for multithreaded applications.
Select one:
1. waiting
2. lightweight
3. heavyweight
4. sleeping

Question 100
Version 6 of the Internet Protocol (IPv6) defines an IP address as a .... number.
Select one:
1. 256-bit
2. 32-bit
3. 64-bit
4. 128-bit

QUESTION BANK ON INTERPROCESS COMMUNICATION & LINUX DISK FILE SYSTEM

INTER-PROCESS COMMUNICATION (CHAPTER 19)

1. Explain what is meant by Inter Process Communication (IPC).
2. State and Explain the six basic mechanisms offered by linux systems to allow Inter Process Communication.
3. Explain the following :
a) Pipes
b) FIFOs
c) Semaphores
d) Message Queues
e) Shared Memory
f) Sockets
4. Explain what is meant by a pipe.
5. Explain the following functions :
a) pipe() b) execve() c) dup2() d) popen() e) pclose() f) pipe_read() g) pipe_write()
6. Describe all the actions essentially performed by the command shell when the command statement ls | more is interpreted and executed after invoking the pipe command. Also clearly indicate the actions performed by the two child processes created.
7. Describe the operations performed by the function popen().
8. One pipe buffer consists of .... page frame(s)
a) 1 b) 4 c) 8 d) 16
9. Each pipe makes use of .... pipe buffers which greatly enhances the performance of User Mode applications that write large chunks of data in a pipe.
a) 1 b) 4 c) 8 d) 16
10. Describe pipefs special filesystem. Indicate where this is located.
11. State the three limitations of a pipe.
12. Explain what is meant by named pipe or FIFO.
13. Distinguish between Pipes & FIFOs.
14. What are the advantages of FIFO over unnamed pipe?

15. Explain the operations performed by the following functions :
    a) mknod() b) mkfifo()
16. Discuss System V IPC (Inter Process Communication) mechanisms
17. List the three kinds of System V IPC resources.
18. Explain the purpose of the following :
    a) IPC key b) IPC Resource identifier
19. Explain the purpose of the following functions :
    a) semget() b) msgget() c) shmget()
20. Explain the method used to compute IPC resource Identifier by providing the expression used and indicating clearly all the parameters used in the expression to compute the IPC identifier.
21. Explain IPC Semaphores.
22. Distinguish between kernel semaphore and IPC semaphore.
23. What are primitive semaphores?
24. Explain the relation between primitive and IPC semaphore
25. Number of IPC semaphore resources that can be used by default is ....
    a) 32 b) 64 c) 128 d) 256
26. Number of primitive semaphores available inside a single IPC semaphore resource by default is .....
    a) 30 b) 60 d) 120 d) 250
27. Describe the fail-safe mechanism used by System V IPC semaphores for situations in which a process dies without being able to undo the operations that it previously issued on a semaphore.
28. Explain what is meant by undoable semaphore operations.
29. What are per-process list and per-semaphore list? Why are they required in semaphore operations?
30. Explain what is meant by IPC message queue.
31. Describe the two functions used by the processes to send and receive messages via the IPC message queue resource.
32. Discuss the following functions clearly explaining the parameters required by each of these functions.
    a) msgsnd() b) msgrcv()
33. In Unix System V,
    a) Number of IPC message queue resources allowed by default is ......
    b) Size of each message in a queue by default is ....
    c) Maximum total size of the messages in a queue by default is ....
34. Describe the IPC shared memory resource.
35. Explain the purpose of the following functions :
    a) shmget() b) shmat() c) shmdt()
36. Explain clearly how swapping out pages of IPC shared memory regions are handled by the kernel.
37. Describe the operations performed by the following library functions for POSIX message queues :
    a) mq_open() b) mq_close() c) mq_unlink() d)mq_send()
    e) mq_timedsend() f) mq_receive() g) mq_timedreceive()

File: on-disk file system (Chapter 18)

1) Discuss general characteristics/features of Extended File System Ext2/Ex3/Ex4
2) Explain how faster symbolic links are supported in ex2/ex4 file systems.
3) Explain what is meant by consistency check of the filesystem. Indicate the command used in linux to perform consistency check of the filesystem.
4) Explain what is meant by an immutable file. Write the command in linux to make a file immutable.
5) Explain what is meant by append-only file. Write the command in linux to make a file append-only.

6) Explain the purpose of the following commands :
   a) lsattr b) chattr c) ls -i d) extundelete
7) State what happens when the following commands are executed :
   a) chattr +i /etc/passwd
   b) chattr +a file1
   c) lsattr /etc/passwd
   d) chattr –R +i dir1
   e) extundelete –Restore-all
8) Explain the purpose of the following features in linux :
   a) Undelete b) Journaling
9) With the help of block diagram, describe how EX2 disk partition is divided into block Groups and also draw the layout of a block group showing all the fields.
10) With the help of block diagram, describe how EX4 disk partition is divided into block Groups and also draw the layout of a block group showing all the fields.
11) State the purpose of "Reserve GDT Block" in the Block group of Ext4 filesystem.
12) Size of the Boot block in Ext4 partition is .... bytes.
13) State the purpose of sparse_super feature flag in Ext4 file system.
14) Distinguish between Ext2 and Ext4 filesystems.
15) Draw the layout of a block group in Ex2/Ex4 partition showing all the fields and providing the purpose of each of the field and number of blocks required by each of the field.
16) Describe the following :
   a) Super block b) Block group descriptor c) datablock bitmap
   d) inode bitmap e) inode f) inode table
17) All fields in Ext4 are written to disk in .... byte order. However, all fields in jbd2 (the journal) are written to disk in .... byte order.
18) State the reason to keep a copy of the superblock and a copy of the group of block group descriptors in each block group?
19) Consider a 32-GB Ext2 partition with a 4-KB block size. Determine the approximate number of blockgroups in that partition.
20) Consider a 512-GB Ext2 partition with a 4-KB block size. Determine the approximate number of blockgroups in that partition.
21) Given the following parameters pertaining to ext2 disk file system :
   'm' - number of Blockgroups in a given disk partition
   'n' - number of disk blocks in each of the blockgroups.
   'b' - size of a disk block in bytes

'p' - number of inodes in a blockgroup

Compute the following :

a) Write a general expression for 'n' i.e., to compute the number of disk blocks required to store one blockgroup, i.e. compute the size of a blockgroup in terms of disk blocks required.

b) If b=disk block size = 4096 bytes (4KB), p=8192 inodes and m = 4K Blockgroups then determine the following

(i) Compute

a) size of superblock,
b) size of blockgroup descriptors,
c) size of data block bitmap,
d) size of inode bitmap,
e) size of inode and size of inode table,
f) number of data blocks and size of data blocks

(ii) Size of one blockgroup, i.e., n

(iii) Size of the disk partition.

22) If disk block size = 1KB, # of inodes = 8192 inodes and 8K Blockgroups then determine the following

a) size of superblock,
b) size of blockgroup descriptors,
c) size of data block bitmap,
d) size of inode bitmap,
e) size of inode and size of inode table,
f) number of data blocks and size of data blocks
g) Size of one blockgroup
h) Size of the disk partition.

23) If disk block size = 2 KB, # of inodes = 8192 inodes and 16K Blockgroups then determine the following:

a) size of superblock,
b) size of blockgroup descriptors,
c) size of data block bitmap,
d) size of inode bitmap,
e) size of inode and size of inode table,
f) number of data blocks and size of data blocks
g) Size of one blockgroup
h) Size of the disk partition.

24) If disk block size = 8 KB, # of inodes = 8192 inodes and 8K Blockgroups then determine the following:

a) size of superblock,
b) size of blockgroup descriptors,
c) size of data block bitmap,
d) size of inode bitmap,
e) size of inode and size of inode table,
f) number of data blocks and size of data blocks
g) Size of one blockgroup
h) Size of the disk partition.

25) If disk block size = 16KB, # of inodes = 8192 inodes and 16K Blockgroups then determine the following:
a) size of superblock,
b) size of blockgroup descriptors,
c) size of data block bitmap,
d) size of inode bitmap,
e) size of inode and size of inode table,
f) number of data blocks and size of data blocks
g) Size of one blockgroup
h) Size of the disk partition.

26) How many block groups are there in a given 512 GB Ext2 partition with a 4-KB block size and 8192 inodes in each block group.
27) Explain what is meant by file extent.
28) Write the various fields in a blockgroup descriptor data structure ext2_group_desc and describe the information provided by each of the field in the blockgroup descriptor data structure.
29) Write the various fields in inode (file descriptor) data structure ext2_inode and describe the information provided by each of the field in the inode data structure.
30) Explain the purpose of following fields in an inode data structure :
a) i_mode, b) i_uid, c) i_size, d) i_links_count,
e) i_blocks, f) i_block[15], g) i_file_acl, h) i_dir_acl, i) i_atime,
j) i_ctime, k) i_mtime, l) i_dtime, m) i_gid
31) What is Access Control List in Ext2/Ex4 FileSystem?
32) Write the linux command to create access control list for individual users or groups for a given file.
33) Explain the following Unix commands by giving the correct syntax and taking some examples
a) setfacl b) getfacl
34) State the reason for including Extended attributes for Inode Objects?
35) Explain how i_file_acl field of the inode is used to create extended attributes of an inode.
36) Describe the layout of an Ex2 extended attributes structure inside a block with the help of a diagram.
37) What is the size of each inode in the inode table? Describe the approach used by Ext4 to increase the size of each inode.
38) What is the maximum size of a file as limited by i_size field of an inode? Describe the approach used by Ext4 to increase the file size considering 64-bit architecture.
39) Given 32-GB Ext2 partition with a 4-KB block size. Assume that all the inodes have the same size: 128 bytes. Data block bitmap and inode bitmap take one block each. Determine the number of blocks occupied by the inode table and size of the inode table.
40) What are the different file types recognized by Ext2 disk file system?
41) Explain the following file types:
a)Regular file (Type 1) b)Directory (Type 2)
c)Character device (Type 3) d)Block device (Type 4)

e)Named pipe (Type 5) f)Socket (Type 6)
g)Symbolic link (Type 7)

42) Indicate the types of files which require no data blocks on the disk.
43) Explain why no data block is required on the disk if symbolic link has upto 60 characters.
44) Describe the two steps used in general to create a filesystem on a disk.
45) Write the command used in linux to create a filesystem on a disk. Indicate the correct syntax of the command by giving an example.

46) Consider the Ex2 Directory entry structure as shown in the table below :

address	inode	rec_len	name_len	file_type	name
0	21			2	.
	22			2	..
	46			1	file1
	53			1	home12
	0			3	kbd1
	67			2	user1
	75			7	file2
	60			4	fd1
	0			1	Oldfile23
	34			2	sbin
	89			6	sock1

a) Identify Regular files, Directories, character devices, block devices, named pipes, sockets and symbolic links
b) Identify the deleted files
c) List the file names requiring \0 padding and also specify how many \0 paddings required.
d) Write the specific values in address, rec_len and name_len fields.

47) Indicate which of the following types of fields in the Ext2 data structures are :
Always cached or stored in Dynamic memory or Never cached :
Super block ....
Group Descriptor ....
Block bitmap ....
Inode bitmap ....
Inode ....
Data block ....
Free inode ....
Free block ....

48) List out all the actions performed by make2fs program.
49) State the purpose of a lost+found directory.
50) State the purpose of the following command :
$ dd if=/dev/sda1 of=/dev/sda6
51) State the operation performed by the following command :
$ dd if=/dev/fd0 bs=1k count=1440 | od -tx1 -Ax > /tmp/dump_hex

52) State the two main problems that must be addressed when the Ext2 filesystem manages the disk space by allocating and deallocating inodes and data blocks.

53) Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the following:
  a) Total number of blocks in the floppy disk considering its size.
  b) Total number of reserved blocks
  c) Number of Block group descriptors
  d) Number of inodes in the inode table
  e) Size of inode table in terms of blocks

54) Describe the two steps used to derive the logical block number of the corresponding data block from an offset f inside a file.

55) Assume a data block size of 4KB, find the file block number for the given file offset 14320. Also specify the corresponding logical block number.

56) Assume a data block size of 1KB, find the file block number for the given file offset 14320. Also specify the corresponding logical block number.

57) Assume a data block size of 4KB, find the file block number for the given file offset 348320. Also specify the corresponding logical block number.

58) Assume a data block size of 1KB, find the file block number for the given file offset 348320. Also specify the corresponding logical block number.

59) Assume a data block size of 4KB, find the file block number for the given file offset 6782341. Also specify the corresponding logical block number.

60) Explain with the help of a schematic, how Data Block Addressing is implemented in Ext2 filesystem.

61) Describe clearly how i_block array field in the disk inode is used to address the disk data blocks pertaining to a file.

62) Determine how many disk accesses are required to access a file data block in
  a) one of the first 12 components of the i_block array.
  b) 13th component of the i_block array.
  c) 14th component of the i_block array
  d) 15th component of the i_block array.

63) With the help of a schematic, explain the following with respect to Data Block addressing considering a block size of 4KB:
  a) (i) What do the first 12 components of the i_block array i.e., i_block[0] to i_block[11] contain?
  (ii) What should be the maximum size of the file that can be addressed by these 12 components?
  b) (i) What does the 13th component of i_block array, i.e., i_block[12] contain? How many levels of indirect addressing is used?
  (ii) How many data blocks pertaining to a file can be addressed by this component? Also specify the starting and ending numbers (indices) of these data block pointers. Explain with the help of a schematic.
  (iii) What should be the maximum size of the file that can be addressed considering only 13th component of the array.

(iv) What should be the maximum size of the file that can be addressed by considering first 13 components of the i_block array?

c) (i) What does the 14th component of i_block array, i.e., i_block[13] contain? How many levels of indirect addressing is used?

(ii) How many data blocks pertaining to a file can be addressed by this component Also specify the starting and ending numbers (indices) of these data block pointers. Explain with the help of a schematic.

(iii) What should be the maximum size of the file that can be addressed considering only 14th component of the array.

(iv) What should be the maximum size of the file that can be addressed by considering first 14 components of the i_block array?

d) (i) What does the 15th component of i_block array, i.e., i_block[14] contain? How many levels of indirect addressing is used?

(ii) How many data blocks pertaining to a file can be addressed by this component. Also specify the starting and ending numbers (indices) of these data block pointers. Explain with the help of a schematic.

(iii) What should be the maximum size of the file that can be addressed by only 15th component of the array.

(iv) What should be the maximum size of the file that can be addressed by considering first 15 components of the i_block array?

e) Determine the total amount of disk blocks required to store all the 15 components of the i_block array i.e.,all direct block pointers and their all levels of indirect block pointers (second order array, third order array, fourth order array, etc).

f) Determine Maximum amount of disk blocks required to store all the block pointers(as computed in e)) and all file data blocks.

64) Repeat all the parts of Q. 59) considering a block size of 1KB.

65) Repeat all the parts of Q. 59) considering a block size of 2KB.

66) Repeat all the parts of Q. 59) considering a block size of 8KB.

67) Repeat all the parts of Q. 59) considering a block size of 64KB.

68) Specify the File size upper limits for data block addressing in the following table :

Block Size	Direct	1-Indirect	2-indirect	3-indirect
1KB				
2KB				
4KB				
8KB				
16KB				

69) Explain what is meant by a file hole. Give an example.

70) Write the file size and total number of blocks assigned to a file hole when the following commands are executed. Also show the i_block array and file /tmp/hole data block contents.
   a) $ echo -n "X" | dd of=/tmp/hole bs=1024 seek=6
   b) $ echo -n "X" | dd of=/tmp/hole bs=4096 seek=6
   c) $ echo "HELLO" | dd of=/tmp/hole bs=1024 seek=10
   d) $ echo "HELLO" | dd of=/tmp/hole bs=4096 seek=10

71) State the purpose of the following commands :
$ mke2fs -t ext4 -b 4096 /dev/sda1
$fsck

72) State the purpose of the following commands :
$ mount -t ext4 /dev/fd0 /floppy
$mount -t ext4 -o data=writeback /dev/sda2 /jdisk

73) State the purpose of the following commands :
$setfacl -m “u:user_name:rwx” file1
$setfacl -m “g:group_name:rwx” file1
$getfacl file1

74) Write a linux command to display only the contents of the superblock of ext4 filesystem stored on the disk partition /dev/sda1.

75) Write a linux command to display only the contents of the block group descriptors of ext4 filesystem stored on the disk partition /dev/sda1.

76) State what happens when the following linux commands are executed. Clearly write the outputs obtained.
   a) $ dumpe2fs -h /dev/sda1
   b) $ dumpe2fs -g /dev/sda1

77) State what happens when the following linux commands are executed. Clearly write the outputs obtained.
   a) $tune2fs –l /dev/sda1
   b) $tune2fs -l /dev/sda1 | grep “Default mount options”
   c) $tune2fs -o acl /dev/sda1

78) State what happens when the following linux commands are executed. Clearly write the outputs obtained.
   a) $ lsblk -O
   b) $ fdisk –l

79) State what happens when the following linux commands are executed. Clearly write the outputs obtained.
   a) $ls –ia
   b) $lscpu
   c) $lsipc

80) Define Journaling and explain why is it required.

81) Explain the idea behind Ext3/4 Journaling to perform each high-level change to the filesystem.

82) Describe how journaling is used to recover from system crashes.

83) Explain the two kinds of blocks in filesystem that need to be logged to the Journal.

84) How does the kernel check whether the filesystem was unmounted properly previously or not?

85) State the purpose of s_mount_state field of the superblock on disk.

86) Explain the three different Journaling modes provided by Ext3/Ext4. State the merits and demerits of each of these modes.

87) Explain the following Journaling modes indicating clearly their advantages and disadvantages.

a) Journal b) ordered c) writeback

88) Explain how Journaling mode of the Ext3/Ext4 filesystem can be specified at the time of mounting the filesystem.

89) Write the linux command to mount an Ext3 filesystem stored in the /dev/sda2 partition on the /jdisk mount point with the "writeback" mode.

90) The Ext3 journal is usually stored in a hidden file named ....located in the root directory of the filesystem.

QUIZ QUESTIONS:

Question 1

A process creates a FIFO by issuing a ....system call passing to it as parameters the pathname of the new FIFO and the value S_IFIFO (0x10000) logically ORed with the permission bit mask of the new file.

Question 2

In Ext2 superblock structure, ... field provides number of blocks per block group

Question 3

In Ext4, If the .... feature flag is set, redundant copies of the superblock and group descriptors are kept only in the groups whose group number is either 0 or a power of 3, 5, or 7.

Question 4

SAY TRUE OR FALSE

"If the kernel appends some data to an existing file so that the number of data blocks allocated for it increases, the values of the s_free_blocks_count field in the Ext2 superblock and of the bg_free_blocks_count field in the group descriptor must be modified."

Question 5

.... is a data structure in a Unix-style file system which describes a filesystem object such as a file or a directory.

Question 6

In Ext2 superblock structure, ... field provides time of last mount operation.

Select one:

1. s_mnttime
2. s_mtime
3. s_wtime
4. s_checkinterval

Question 7
.... field in inode object provides the inode number

Question 8
The most important field in the sem_queue data structure for a pending request is .... which is a pointer to an array of integer values describing each pending semaphore operation,
Select one:
1. sops
2. sma
3. status
4. nsops

Question 9
For a given 64-GB Ext4 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 256
2. 512
3. 4096
4. 1024

Question 10
The ....includes all sem_undo data structures corresponding to IPC semaphores on which the process has performed undoable operations.

Question 11
In ext4, by default, a filesystem is allowed to increase in size by a factor of ....x over the original filesystem size.

Question 12
" Memory regions used for IPC shared memory always define the nopage method. "
Say True or False

Question 13
SAY TRUE OR FALSE
"When creating an Ext2 filesystem, the system administrator may choose the optimal block size (from 1,024 to 4,096 bytes), depending on the expected average file length".

Question 14
In Ext2 Block group descriptor structure, ... field provides Block number of inode bitmap.

Question 15
.... field in superblock object provides list of inodes waiting to be written to disk

Question 16
For a given 32-GB Ext2 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 8192
2. 4096
3. 256
4. 1024

Question 17
SAY TRUE OR FALSE
"When a new file is created, the values of the s_free_inodes_count field in the Ext2 superblock and of the bg_free_inodes_count field in the proper group descriptor must be decreased."
Question 18
"To balance the number of regular files and directories in a block group, Ext2 introduces a debt parameter for every block group. "
SAY TRUE OR FALSE

Question 19
If disk block size = 16KB, # of inodes = 32768 inodes and Partition size = 512GB then determine the Number of blockgroups in the file system
Select one:
1. 1024
2. 512
3. 256
4. 128

Question 20
.................field in msg_queue data structure provides number of bytes in queue
Select one:
1. q_qbytes
2. q_qcbytes
3. q_messages
4. q_qnum

Question 21
.... sets Access Control Lists (ACLs) of files and directories.

Question 22
Suppose that each block group contains 4,096 inodes, the address on disk of inode 13,021 can be given as ....
Select one:
1. Blockgroup 3, 13021
2. Blockgroup 0, 13021
3. Blockgroup 3, 733
4. Blockgroup 2, 733

Question 23
.................field in msg_queue data structure provides list of messages in queue
Select one:
1. q_messages
2. q_qbytes
3. q_qnum
4. q_qcbytes

Question 24
In Ext2 Block group descriptor structure, ... field provides Number of free blocks in the group.

Question 25
No data blocks are required for one of these kinds of files.
Select one:
1. Symbolic Link
2. Device file
3. Directory
4. Regular file

Question 26
In .... journaling mode, Ext3 filesystem groups metadata and relative data blocks so that data blocks are written to disk before the metadata.

Question 27
In Ext4 file system, Journal size is ....
Select one:
1. 64MB
2. 256MB
3. 512MB
4. 128MB

Question 28
Consider Data Block addressing in Ext2 disk file system with a block size of 4KB. Determine the maximum size of the file that can be addressed by first 12 components of the i_block array.
Select one:
1. 48KB
2. 4MB
3. 4GB
4. 4TB

Question 29
For a given 32-GB Ext2 partition with a 1-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 4096
2. 8192
3. 256
4. 1024

Question 30
............Best suited to implement producer/consumer interactions among processes.
Select one:
1. Semaphore
2. Pipe/FIFO
3. Message Queues
4. Shared Memory

Question 31
Determine how many disk accesses (worst case) are required to access a file data block in 14th component i_block[14] of the i_block array.
Select one:
1. 6
2. 5
3. 4
4. 3

Question 32
In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then the number of data blocks available for storing file data is .... blocks
Select one:
1. 1024
2. 8192
3. 7921
4. 4355

Question 33
For given a 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192, then the total number of blocks in a blockgroup is .... blocks.
Select one:
1. 2048
2. 16384
3. 32768
4. 65536

Question 34
For a given 512-GB Ext4 partition with a 8-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 1024
2. 4096
3. 512
4. 256

Question 35
A process using IPC message queue resource invokes the .... function to receive a message.

Question 36
In Ext2 Block group descriptor structure, ... field provides Block number of block bitmap.

Question 37
In Unix command shells, ... can be created by means of the | operator.
Select one:
1. FIFO
2. Message Queues
3. Semaphore
4. Pipe

Question 38
In .... journaling mode, all filesystem data and metadata changes are logged into the journal.

Question 39
In ext2_inode structure, ... field provides file type and access rights
Select one:
1. i_size
2. i_blocks
3. i_file_acl
4. i_mode

Question 40
If the symbolic link represents a short pathname with at most .... characters, it can be stored in the inode and can thus be translated without reading a data block.
Select one:
1. 256
2. 60
3. 128
4. 128

Question 41
Assume a data block size of 1KB, if the file block number is 340 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[13], first level indirectblock[0], second level indirectblock0[72]
2. i_block[13], first level indirectblock[0] & second level indirectblock0[85]
3. i_block[12] & first level indirectblock[72]
4. i_block[14], first level indirectblock[1] & second level indirectblock0[72]

Question 42
"Pages of an IPC shared memory region are swappable and not syncable because they map a special inode that has no image on disk."
Say True or False

Question 43
State whether the following statement is TRUE or FALSE.
“In most traditional Unix-like kernels, each filesystem can be mounted only once. However, in Linux - it is possible to mount the same filesystem several times.” ....

Question 44
In a pipe_inode_info structure, ... field provides Number of buffers containing data to be read.

Question 45
SAY TRUE OR FALSE
"When creating an Ext2 filesystem, the system administrator may choose how many inodes to allow for a partition of a given size, depending on the expected number of files to be stored on it."

Question 46
.... command is used to create a Ext2 filesystem on a hard disk
Select one:
1. e2fsck
2. tune2fs
3. mke2fs
4. dumpe2fs

Question 47
Full form of Ext2 is ....

Question 48
The Character device file is associated with a file type number ....

Question 49
In Ext2 superblock structure, ... field provides Block group number of this superblock.
Select one:
1. s_blocks_count
2. s_r_blocks_count
3. s_blocks_per_group
4. s_block_group_nr

Question 50
In a pipe_inode_info structure, ... field provides Read position in current pipe buffer.

Question 51
The ....includes all sem_undo data structures corresponding to the processes that performed undoable operations on the semaphore.

Question 52
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 and Partition size = 256GB then determine the Total number of blockgroups.
Select one:
1. 32786
2. 8192
3. 4096
4. 512

Question 53
In Ext2 superblock structure, ... field provides free inodes count in the filesystem.

Question 54
Each pipe makes use of .... pipe buffers which greatly enhances the performance of User Mode applications that write large chunks of data in a pipe.
Select one:
1. 1
2. 8
3. 16
4. 4

Question 55
Assume a data block size of 4KB, find the file block number for the given file offset 348320.
Select one:
1. 85
2. 105
3. 75
4. 95

Question 56
For a given Ext2 file system having 8192 inodes in each blockgroup with blocksize 4KB, the total number of blocks required by the Inode Table is ....
Select one:
1. 512
2. 1024
3. 128
4. 256

Question 57
Actually, in the 80 x 86 architecture, there is just one IPC system call named ....

Question 58
In Unix System V, Size of each message in a queue by default is ....bytes

Question 59
For given a 64GB partition having Ext4 file system with disk block size = 8KB, # of inodes = 8192 inodes, then the total number of blocks in a blockgroup is .... blocks
Select one:
1. 32768
2. 16384
3. 2048
4. 65536

Question 60
A pipe is implemented as a set of VFS objects, which have no corresponding disk images.
Say True or False
Question 61
Write a single UNIX command to replace the following two commands :

$ ls > temp
$ more < temp

Question 62
In ext2_inode structure, ... field provides number of data blocks of the file in terms of sectors.
Select one:
1. i_file_acl
2. i_block
3. i_size
4. i_blocks
Question 63
Each IPC resource also has a ....-bit IPC identifier, which is somewhat similar to the file descriptor associated with an open file.
Select one:
1. 16
2. 64
3. 32
4. 128
Question 64
When two or more processes wish to communicate through an IPC resource, they all refer to the IPC identifier of the resource.
Say True or False

Question 65
Assume a data block size of 4KB, if the file block number is 1655 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[12] & first level indirectblock[619]
2. i_block[13], first level indirectblock[0] & second level indirectblock0[619]
3. i_block[13], first level indirectblock[1] & second level indirectblock0[1655]
4. i_block[14], first level indirectblock[1] & second level indirectblock0[72]

Question 66
Assume a data block size of 1KB, if the file block number is 6623 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[13], first level indirectblock[0] & second level indirectblock24[211]
2. i_block[13], first level indirectblock[1] & second level indirectblock25[211]
3. i_block[14], first level indirectblock[0] & second level indirectblock0[25]
4. i_block[12] & first level indirectblock[255]

Question 67
One pipe buffer consists of .... page frame(s)
Select one:
1. 8
2. 16
3. 1
4. 4

Question 68
Given Ex2 filesystem with default data block size=4KB, write the i_size and i_blocks fields for a file hole when the following command is executed.

```
$ echo -n "X" | dd of=/tmp/hole bs=1024 seek=6
```

Select one:
1. i_size = 6144, i_blocks =1
2. i_size = 6145, i_blocks =8
3. i_size = 6145, i_blocks =1
4. i_size = 6144, i_blocks =8

Question 69
The following command displays the detailed group information block numbers in hexadecimal format.
Select one:
1. $dumpe2fs -b /dev/sda1
2. $dumpe2fs -f /dev/sda1
3. $dumpe2fs -x /dev/sda1
4. $dumpe2fs -h /dev/sda1

Question 70
When shell executes the following command, ... process takes the input from first pipe and writes the output to the second pipe

```
$ head -60 tfile | tail -1 | wc -c
```

Select one:
1. head -60 tfile
2. wc -c
3. tail -1
4. shell

Question 71
The size of each inode in the inode table in ext4 is ....bytes
Select one:
1. 128
2. 64
3. 32
4. 256

Question 72
POSIX introduces a function named ....specifically to create a FIFO.

Question 73
How many processes are created by the shell when it executes the following command :
$ ls | more
Select one:
1. 2
2. 3
3. 1
4. 4

Question 74
If the pathname of a symbolic link is longer than .... characters, a single data block is required

Question 75
Assume a data block size of 1KB, find the file block number for the given file offset 6782341.
Select one:
1. 6523
2. 6423
3. 6723
4. 6623

Question 76
Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB. Determine the maximum size of the file that can be addressed by 12th component of the i_block array.
Select one:
1. 48KB
2. 4MB
3. 4GB
4. 4TB

Question 77
The Unknown file is associated with a file type number ....

Question 78
Each IPC shared memory region is associated with a file belonging to the ....special filesystem located in VFS layer in Linux

Question 79
.... command with option .... is used to make a file append-only.
Select one:
1. chattr -a
2. chattr +a
3. chattr +i
4. chattr -i

Question 80
For a given 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of data blocks available for storing file data is .... blocks
Select one:
1. 32768
2. 259
3. 32506
4. 771

Question 81
SAY TRUE OR FALSE
"In Ext2 filesystem, both the superblock and the group descriptors are always duplicated in each block group."

Question 82
Many Unix systems provide, besides the pipe( ) system call, two wrapper functions named .... & ....that handle all the work usually done when using pipes.
Select one or more:
1. pipe()
2. popen()
3. close()
4. pclose()

Question 83
In Ext2 superblock structure, ... field provides filesystem size in blocks.
Select one:
1. s_blocks_count
2. s_blocks_per_group
3. s_log_block_size
4. s_free_blocks_count

Question 84
If the pathname of a symbolic link has up to .... characters, it is stored in the i_block field of the inode.

Question 85
Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the Total number of reserved blocks considering the exact number of blocks on the floppy.
Select one:
1. 83
2. 73
3. 70
4. 80

Question 86
A Page Fault occurs when a process tries to access a location of an IPC shared memory region whose underlying page frame has not been assigned. The corresponding exception handler determines that the faulty address is inside the process address space and that the corresponding Page Table entry is null; therefore, it invokes the .... function.

Question 87
..................function establishes an asynchronous notification mechanism for the arrival of messages in an empty POSIX message queue
Select one:
1. mq_close()
2. mq_notify( )
3. mq_open( )
4. mq_unlink( )

Question 88
Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB. Determine the maximum size of the file that can be addressed by 13th component of the i_block array.
Select one:
1. 4MB
2. 4TB
3. 48KB
4. 4GB

Question 89
In ext2_inode structure, ... field provides Time of last file access.
Select one:
1. i_atime
2. i_dtime
3. i_ctime
4. i_mtime

Question 90
The number of blocks required to store Inode Bitmap in a Block group is ....

Question 91
If disk block size = 32KB, # of inodes = 32768 and Partition size = 512GB then determine the Size of one blockgroup in the file system.
Select one:
1. 4GB
2. 8GB
3. 2GB
4. 512MB

Question 92
Number of processes created by the shell when it executes the following command is ....

$ head -60 tfile | tail -1 | wc -c

Question 93
For a given 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the size of one blockgroup is .... blocks
Select one:
1. 2048MB
2. 128MB
3. 1024MB
4. 512MB

Question 94
For a given 512-GB Ext4 partition with a 4-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 2048
2. 4096
3. 512
4. 1024

Question 95
In Ext2 Block group descriptor structure, ... field provides Block number of first inode table block.

Question 96
System V IPC first appeared in a development Unix variant called ....
Select one:
1. Columbus Unix
2. Linux
3. Sun Solaris
4. AT&T's System III

Question 97
..................field in msg_queue data structure provides number of messages in queue
Select one:
1. q_qcbytes
2. q_qnum
3. q_qbytes
4. q_messages

Question 98
For given a 64GB partition having Ext2 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the total number of blockgroups is .... blocks
Select one:
1. 2048
2. 512
3. 256
4. 1024

Question 99
For a given Ext4 file system having 8192 inodes in each blockgroup with blocksize 4KB, the total number of blocks required by the Inode Table is ....
Select one:
1. 128
2. 512
3. 1024
4. 256

Question 100
For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the total number of blockgroups is .... blocks
Select one:
1. 1024
2. 2048
3. 256
4. 512

Question 101
In Ext4 file system, a 1,024-byte block contains .... inodes, while a 4,096-byte block contains ..... inodes.
Select one:
1. 4, 32
2. 4, 16
3. 8, 32
4. 8, 64

Question 102
Each ....is a set of one or more semaphore values, not just a single value like a kernel semaphore.

Question 103
SAY TRUE OR FALSE
"To minimize the impact of system crashes, when creating a new hard link for a file, the new name is added into the proper directory first and the counter of hard links in the disk inode is increased next."

Question 104
The size of each inode in the inode table in ext2 is ....bytes.
Select one:
1. 64
2. 32
3. 128
4. 256

Question 105
The following command displays a filesystem – both superblock and all of the block group descriptor detail information.
Select one:
1. $dumpe2fs -x /dev/sda1
2. $dumpe2fs -f /dev/sda1
3. $dumpe2fs -b /dev/sda1
4. $dumpe2fs -h /dev/sda1

Question 106
In Ext2 superblock structure, ... field provides time of last write operation.
Select one:
1. s_mtime
2. s_mnttime
3. s_checkinterval
4. s_wtime

Question 107
In Ext4 filesystem, when sb.s_log_block_size =2, Block size = ...
Select one:
1. 64KB
2. 8KB
3. 4KB
4. 1KB

Question 108
.....................function closes a POSIX message queue without destroying it
Select one:
1. mq_notify( )
2. mq_setattr( )
3. mq_unlink( )
4. mq_close()

Question 109
In a pipe_inode_info structure, ... field provides number of reading processes.

Question 110
Suppose that each block group contains 8,192 inodes, the address on disk of inode 23,146 can be given as ....
Select one:
1. Blockgroup 3, 6762
2. Blockgroup 1, 6762
3. Blockgroup 2, 6762
4. Blockgroup 2, 6761

Question 111
Determine how many disk accesses (worst case) are required to access a file data block in 13th component i_block[13] of the i_block array
Select one:
1. 7
2. 4
3. 5
4. 6

Question 112
" It is impossible for two arbitrary processes to share the same pipe, unless the pipe was created by a common ancestor process. "
Say True or false

Question 113
Assume a data block size of 4KB, if the file block number is 3 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[3]
2. i_block[12] & first level indirectblock[3]
3. i_block[13] & first level indirectblock[3]
4. i_block[12] & first level indirectblock[2]

Question 114
In Ext4 file system, Flex Block group size (number of Blockgroups combined in a flex Block group) is ....

Question 115
The Directory file is associated with a file type number ....

Question 116
In ext2 i_node structure, ... field provides Hard links counter.

Question 117
In a pipe_inode_info structure, ... field provides number of writing processes.

Question 118
The ext2_get_block() function handles the data structures and when necessary, invokes the .... function to actually search for a free block in the Ext2 partition.
Select one:
1. ext2_free_inode( )
2. ext2_new_inode( )
3. ex2_get_block()
4. ext2_alloc_block( )

Question 119
A process using IPC message queue resource invokes the .... . function to send a message.

Question 120
The Named pipe (FIFO) file is associated with a file type number ....

Question 121
.......................fields in the Ext2 file system data structures are always cached.
Select one or more:
1. Block Bitmap
2. Block Group Descriptors
3. Inode Table
4. Inode Bitmap
5. Superblock

Question 122
In Ext2 file system, a 1,024-byte block contains .... inodes, while a 4,096-byte block contains .... inodes.
Select one:
1. 8, 64
2. 8, 32
3. 4, 32
4. 4, 16
Question 123
Full form of NTFS is ....

Question 124
Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the number of inodes in the inode table considering the exact number of blocks on the floppy.
Select one:
1. 174
2. 184
3. 194
4. 180
Question 125
Assume a data block size of 4KB, find the file block number for the given file offset 14320.
Select one:
1. 6
2. 4
3. 5
4. 3

Question 126
If disk block size = 16KB, # of inodes = 32768 and Partition size = 512GB then determine the Size of the inode table in blocks in each of the block group in the Ext4 file system.
Select one:
1. 1024
2. 2048
3. 512
4. 256
Question 127
The Ext3 journal is usually stored in a hidden file named ....located in the root directory of the filesystem.
Question 128
For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of blocks required to store Inode table in a block group is .... blocks
Select one:
1. 256
2. 512
3. 128
4. 1024
Question 129
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Size of Inode Table in blocks in each blockgroup.
Select one:
1. 128 blocks
2. 1024 blocks
3. 256 blocks
4. 512 blocks
Question 130
The Symbolic link file is associated with a file type number ....
Question 131
When the e2fsck program executes a consistency check on the filesystem status, it refers to the superblock and the block group descriptors stored in block group -----, and then copies them into all other block groups.
Select one:
1. 5
2. 0
3. 3
4. 7
Question 132
Assume a data block size of 1KB, find the file block number for the given file offset 348320.
Select one:
1. 440
2. 240
3. 340
4. 540

Question 133
Assume a data block size of 1KB, find the file block number for the given file offset 14320.
Select one:
1. 15
2. 19
3. 16
4. 13

Question 134
Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the size of inode table in terms of blocks
Select one:
1. 33
2. 23
3. 20
4. 30

Question 135
.....................function destroys a POSIX message queue
Select one:
1. mq_notify( )
2. mq_close()
3. mq_unlink( )
4. mq_open( )

Question 136
...........................fields in the Ext2 file system data structures are never cached.
Select one or more:
1. Superblock
2. Inode Bitmap
3. Free block
4. Free inode
5. Block Bitmap

Question 137
When a process "attaches" a segment, the kernel invokes .... and creates a new shared memory mapping of the file in the address space of the process.

Question 138
Given Ex2 filesystem with default data block size=4KB, write the file size in bytes and total number of blocks assigned to a file hole when the following command is executed.

```
$ echo -n  "XYZ"  |   dd of=/tmp/hole    bs=1024   seek=4
```

Select one:
1. File Size = 3, block required=1
2. File Size = 4099, block required=5
3. File Size = 4099, block required=1
4. File Size = 4096, block required=5

Question 139
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the total number of blocks a in each Blockgroup.
Select one:
1. 65535
2. 32767
3. 32768
4. 65536

Question 140
SAY TRUE OR FALSE
"File fragmentation increases the average time of sequential read operations on the files, because the disk heads must be frequently repositioned during the read operation."

Question 141
The message text starts right after the msg_msg descriptor; if the message is longer than ....bytes (the page size minus the size of the msg_msg descriptor), it continues on another page, whose address is stored in the next field of the msg_msg descriptor.
Select one:
1. 4072
2. 4000
3. 3072
4. 4096

Question 142
Consider Data Block addressing in Ext2 disk file system with a block size of b bytes. Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 13th component i_block[13] of the i_block array
Select one:
1. (b/4)^2 + (b/4) +12 to (b/4)^3 + (b/4)^2 + (b/4) +11
2. 0 to 11
3. (b/4) + 12 to (b/4)^2 + (b/4) +11
4. 12 to (b/4) + 11

Question 143
In Unix System V, Maximum total size of the messages in a queue by default is ....bytes

Question 144
A file's inode number can be found using the .... command.

Question 145
In ext2_inode structure, ... field provides Time of file deletion.
Select one:
1. i_atime
2. i_mtime
3. i_dtime
4. i_ctime

Question 146
.... field in shmid_kernel data structure provides number of current attaches
Select one:
1. shm_atim
2. shm_nattch
3. shm_segsz
4. shm_file
Question 147
.... function is invoked to create IPC Shared memory resource.
Select one:
1. semget()
2. msgget()
3. shmget()
4. shmat()

Question 148
If disk block size = 16KB, # of inodes = 32768 and Partition size = 512GB then determine the Number of Blocks in a blockgroup in the file system.
Select one:
1. 131072
2. 16384
3. 65536
4. 32786
Question 149
sudo .... /etc is used to make a whole directory (e.g., /etc) including all its content immutable at once recursively,
Select one:
1. chattr +R +i
2. chattr +R -i
3. chattr -R +i
4. chattr -R -i

Question 150
Space management must be .... ; that is, the kernel should be able to quickly derive from a file offset the corresponding logical block number in the Ext2 partition.

Question 151
.... avoids the time-consuming check that is automatically performed on a filesystem when it is abruptly unmounted for instance, as a consequence of a system crash.

Question 152
The Regular file is associated with a file type number ....

Question 153
The Socket file is associated with a file type number ....

Question 154
Determine how many disk accesses (worst case) are required to access a file data block in 12th component i_block[12] of the i_block array.
Select one:
1. 3
2. 4
3. 2
4. 5

Question 155
The .... field of the inode, which is not used for regular files, represents a 32-bit extension of the i_size field.

Question 156
No data blocks are required for one of these kinds of files.
Select one:
1. Socket
2. Directory
3. Symbolic Link
4. Regular file

Question 157
..........................are counters used to provide controlled access to shared data structures for multiple processes.
Select one:
1. IPC Shared Memory
2. IPC semaphores
3. IPC Message Queues
4. Kernel Semaphore

Question 158
The counters inside an IPC semaphore are referred to as ....
Question 159
The per-semaphore list is mainly used when a process invokes the ....function to force an explicit value into a primitive semaphore.
Question 160
In a pipe_inode_info structure, ... field provides a pointer to Pipe/FIFO wait queue
Question 161
When a process chooses to use fail-safe mechanism, the resulting operations are called ....semaphore operations.
Select one:
1. doable
2. kernel
3. primitive
4. undoable

Question 162
For given a 64GB partition having Ext4 file system with disk block size = 4KB, # of inodes = 8192 inodes, then the number of data blocks available for storing file data is .... blocks
Select one:
1. 31229
2. 32768
3. 30009
4. 30229
Question 163
The ....function, which is invoked by do_exit( ), walks through the list and reverses the effect of any unbalanced operation for every IPC semaphore touched by the process.

Question 164
"The goal of a journaling filesystem is to avoid running time-consuming consistency checks on the whole filesystem by looking instead in a special disk area that contains the most recent disk write operations named journal."
Say TRUE OR FALSE
Question 165
In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 inodes and 512 Blockgroups, then total number of disk blocks required to store Block Group descriptors in a block group is .... blocks
Select one:
1. 3
2. 6
3. 24
4. 12

Question 166
The .... function deletes a disk inode, which is identified by an inode object whose address inode is passed as the parameter.
Select one:
1. ext2_free_inode( )
2. ext2_new_inode( )
3. ext2_new_inode( )
4. ext2_free_blocks()

Question 167
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Number of blockgroups
Select one:
1. 256
2. 512
3. 128
4. 1024

Question 168
Number of primitive semaphores available inside a single IPC semaphore resource by default is ....
Select one:
1. 250
2. 30
3. 128
4. 256

Question 169
When the inode object refers to a pipe, its .... field points to a pipe_inode_info structure.

Question 170
---------------- field in msg_queue data structure provides maximum number of bytes in queue
Select one:
1. q_qcbytes
2. q_qnum
3. q_qbytes
4. q_messages

Question 171
.... command with option .... is used to make a file immutable
Select one:
1. chattr +a
2. chattr +i
3. chattr -a
4. chattr -i

Question 172
.... field in shmid_kernel data structure provides segment size in bytes
Select one:
1. shm_nattch
2. shm_segsz
3. shm_atim
4. shm_file

Question 173
Suppose that each block group contains 4,096 inodes, the address on disk of inode 26,042 can be given as ....
Select one:
1. Blockgroup 6, 13021
2. Blockgroup 7, 26042
3. Blockgroup 7, 1467
4. Blockgroup 6, 1466

Question 174
Select the basic mechanisms offered by Linux to allow inter-process communication:
Select one:
1. Pipes and FIFOs (named pipes)
2. Semaphores
3. Messages
4. Shared memory regions
5. Sockets
6. All of the above

Question 175
The .... function creates an Ext2 disk inode, returning the address of the corresponding inode object.
Select one:
1. ext2_new_inode( )
2. ext2_free_inode( )
3. ex2_get_block()
4. ext2_free_blocks()

Question 176
When shell executes the following command, ... process writes the output to the display(screen)
  $ head -60 tfile | tail -1 | wc -c
Select one:
1. tail -1
2. wc -c
3. head -60 tfile
4. shell

Question 177
Determine how many disk accesses (worst case) are required to access a file data block in one of the first 12 components i_block[0] to [11] of the i_block array.
Select one:
1. 5
2. 2
3. 4
4. 3

Question 178
In Ext2 superblock structure, ... field provides 128-bit filesystem identifier.

Question 179
In order to compute the IPC Identifier using the expression IPC Identifier = s * M + i, Linux uses the value of M set to .... where s starts from 0 and 0<= i < M.
Select one:
1. 32768
2. 65536
3. 16384
4. 32767

Question 180
……………… field in inode object provides Pointers for the list of inodes of the superblock.

Question 181
…………………….. function is invoked to create IPC Message Queue resource.
Select one:
1. shmat()
2. shmget()
3. msgget()
4. semget()

Question 182
Assume a data block size of 4KB, if the file block number is 85 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[13], first level indirectblock[85] & second level indirectblock1[73]
2. i_block[13], first level indirectblock[73] & second level indirectblock0[85]
3. i_block[12] & first level indirectblock[73]
4. i_block[12] & first level indirectblock[85]

Question 183
In Unix System V, Number of IPC message queue resources allowed by default is ……

Question 184
Consider Data Block addressing in Ext2 disk file system with a block size of b bytes. Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by first 12 components i.e., i_block[0] to i_block[11] of the i_block array
Select one:
1. (b/4) + 12 to (b/4)^2 + (b/4) +11
2. (b/4)^2 + (b/4) +12 to (b/4)^3 + (b/4)^2 + (b/4) +11
3. 12 to (b/4) + 11
4. 0 to 11

Question 185
The Block device file is associated with a file type number ……………

Question 186
In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then total number of blocks in a block group is …………….. blocks.
Select one:
1. 8192
2. 512
3. 2048
4. 1024

Question 187
In Ext2 Block group descriptor structure, ... field provides Number of directories in the group

Question 188
A file .... is a portion of a regular file that contains null characters and is not stored in any data block on disk.

Question 189
.... field in shmid_kernel data structure provides a pointer to the special file of the segment
Select one:
1. shm_nattch
2. shm_segsz
3. shm_atim
4. shm_file

Question 190
In ext2_inode structure, ... field provides Time that file contents last changed
Select one:
1. i_mtime
2. i_atime
3. i_dtime
4. i_ctime

Question 191
In Ext2 superblock structure, ... field provides free blocks count in the flesystem.

Question 192
Files larger than .... must be opened on 32-bit architectures by specifying the O_LARGEFILE opening flag.
Select one:
1. 256KB
2. 512KB
3. 2 GB
4. 1GB

Question 193
...... inode number in a directory entry of a directory file indicates that the file specified in that entry has been deleted.

Question 194
In ext2_inode structure, ... field provides pointers to data blocks.
Select one:
1. i_block
2. i_size
3. i_file_acl
4. i_blocks

Question 195
In Ext4 file system, Journal length is ....
Select one:
1. 4096 blocks
2. 16384 blocks
3. 8192 blocks
4. 32768 blocks
Question 196
"In fact, metadata's log records are sufficient to restore the consistency of the on-disk filesystem data structures."
SAY TRUE OR FALSE
Question 197
A process creates a new pipe by means of the ....system call.
Question 198
.... is a special filesystem used in linux which is located in VFS layer in memory to expedite the handling of pipes.
Question 199
Assume a data block size of 1KB, if the file block number is 340 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[13], first level indirectblock[0] & second level indirectblock0[85]
2. i_block[12] & first level indirectblock[72]
3. i_block[14], first level indirectblock[1] & second level indirectblock0[72]
4. i_block[13], first level indirectblock[0], second level indirectblock0[72]
Question 200
.... field in superblock object provides List of file objects

Question 201
The .... function is invoked to "attach" an IPC shared memory region to a process.

Question 202
The ....function is invoked to "detach" an IPC shared memory region specified by its IPC identifier that is, to remove the corresponding memory region from the process's address space.
Question 203
All fields in Ext4 are written to disk in ....
Select one:
1. only unsigned form
2. little-endian order
3. only signed form
4. big-endian order

Question 204
..........................fields in the Ext2 file system data structures are cached only in Dynamic memory.
Select one or more:
1. Data block
2. inode
3. Block Bitmap
4. Inode Bitmap
5. Block Group Descriptors
6. Superblock
7. Free block
8. Free inode
Question 205
Number of IPC semaphore resources that can be used by default is ....
Select one:
1. 64
2. 256
3. 32
4. 128

Question 206
In Ext2 Block group descriptor structure, ... field provides Number of free inodes in the group

Question 207
Pipes are fully integrated into the ...... layer, and the kernel can handle them in the same way as named pipes or FIFOs, which truly exist as files recognizable to end users.

Question 208
Given Ex2 filesystem with default data block size=4KB, write the file size and total number of blocks assigned to a file hole when the following command is executed.

```
$ echo -n "XYZ" | dd of=/tmp/hole bs=4096 seek=7
```

Select one:
1. File Size = 28675, block required=1
2. File Size = 28672, block required=7
3. File Size = 3, block required=1
4. File Size = 28675, block required=8

Question 209
When the kernel has to locate a block holding data for an Ext2 regular file, it invokes the .... function. If the block does not exist, the function automatically allocates the block to the file.
Select one:
1. ex2_get_block()
2. ext2_free_blocks()
3. ext2_new_inode( )
4. ext2_free_inode( )

Question 210
When shell executes the following command, ... process takes no inputs from the pipes

$ head -60 tfile | tail -1 | wc -c

Select one:
1. wc -c
2. tail -1
3. shell
4. head -60 tfile

Question 211
In Ext2 superblock structure, ... field provides number of blocks to preallocate for directories

Question 212
In Ext2 superblock structure, ... field provides number of inodes per block group

Question 213
SAY TRUE OR FALSE
"In Ext4 filesystem, both the superblock and the group descriptors are always duplicated in each block group."

Question 214
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the number of Data blocks available for storing File contents in each Blockgroup.
Select one:
1. 64998
2. 32153
3. 63997
4. 65018

Question 215
Consider Data Block addressing in Ext2 disk file system with a block size of b bytes. Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 14th component i_block[14] of the i_block array
Select one:
1. 12 to (b/4) + 11
2. (b/4)^2 + (b/4) +12 to (b/4)^3 + (b/4)^2 + (b/4) +11
3. (b/4) + 12 to (b/4)^2 + (b/4) +11
4. 0 to 11

Question 216
In ext2_inode structure, ... field provides Time that inode last changed.
Select one:
1. i_atime
2. i_mtime
3. i_dtime
4. i_ctime

Question 217
The following command displays the blocks which are reserved as bad in the filesystem.
Select one:
1. $dumpe2fs -h /dev/sda1
2. $dumpe2fs -f /dev/sda1
3. $dumpe2fs -x /dev/sda1
4. $dumpe2fs -b /dev/sda1

Question 218
.... field in shmid_kernel data structure provides PID of last accessing process
Select one:
1. shm_atim
2. shm_nattch
3. shm_lprid
4. shm_cprid

Question 219
.... function creates a POSIX message queue
Select one:
1. mq_unlink( )
2. mq_open( )
3. mq_close( )
4. mq_notify( )

Question 220
A .... is a one-way flow of data between processes.
Select one:
1. FIFO
2. Message Queues
3. Semaphore
4. Pipe

Question 221
The .... field in the sem_array data structure points to an array of sem data structures, one for every IPC primitive semaphore.
Select one:
1. sops
2. sem_base
3. sma
4. status

Question 222
"The Ext2 filesystem status is stored in the s_mount_state field of the superblock on disk."
Say TRUE OR FALSE.

Question 223
The .... field of an inode points to the block containing the extended attributes.

Question 224
In Ext4 filesystem, if disk block size = 8 KB, # of inodes = 16384 inodes and Partition size = 256GB then determine the Size of one blockgroup.
Select one:
1. 256MB
2. 128MB
3. 512MB
4. 1024MB

Question 225
No data blocks are required for one of these kinds of files.
Select one:
1. Named pipe
2. Symbolic Link
3. Regular file
4. Directory

Question 226
.... function is invoked to create IPC Semaphore resource.
Select one:
1. semget()
2. msgget()
3. shmget()
4. shmat()

Question 227
In Ext2 superblock structure, ... field provides total number of inodes in the filesystem
Select one:
1. s_inodes_count
2. s_inodes_per_group
3. s_inode_size
4. s_free_inodes_count

Question 228
Number of pipes created by the shell when it executes the following command is ......

echo " WINE Is Not windows Emulator" | rev | cut –f1 –d' ' | rev | wc -c

Question 229
For each pipe, the kernel creates .... inode object plus .... file objects
Select one:
1. 2, 2
2. 2, 4
3. 1, 4
4. 1, 2

Question 230
In ext2_inode structure, ... field provides file length in bytes
Select one:
1. i_block
2. i_file_acl
3. i_size
4. i_blocks

Question 231
.... command is used to perform automatic consistency checks on the filesystem status
Select one:
1. mkfs
2. e2fsck
3. tune2fs
4. dumpe2fs

Question 232
Consider Data Block addressing in Ext2 disk file system with a block size of b bytes. Determine the range of file block numbers of data blocks (i.e., number or index of the data blocks) that can be addressed by 12th component i_block[12] of the i_block array
Select one:
1. 0 to 11
2. (b/4) + 12 to (b/4)^2 + (b/4) +11
3. 12 to (b/4) + 11
4. (b/4)^2 + (b/4) +12 to (b/4)^3 + (b/4)^2 + (b/4) +11

Question 233
The number of blocks required to store Data Block Bitmap in a Block Group is ....

Question 234
Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the Total number of blocks (EXACT) in the floppy disk considering its size.
Select one:
1. 1574
2. 1400
3. 1474
4. 1500

Question 235
Assume a data block size of 4KB, find the file block number for the given file offset 6782341.
Select one:
1. 1855
2. 1655
3. 1755
4. 1555

Question 236
For a given 64-GB Ext4 partition with an 8-KB block size, the approximate number of Block Groups in the disk partition is ....
Select one:
1. 256
2. 128
3. 4096
4. 1024

Question 237
In Ext2 superblock structure, ... field provides number of blocks to preallocate for files

Question 238
The number of bytes currently written in all pipe buffers and yet to be read is called ....

Question 239
.............Allow processes on different computers to exchange data through a network.
Select one:
1. Semaphore
2. Sockets
3. Message Queues
4. Shared Memory

Question 240
Consider Data Block addressing in Ext2/ext4 disk file system with a block size of 4KB. Determine the maximum size of the file that can be addressed by 14th component of the i_block array.
Select one:
1. 4GB
2. 4TB
3. 4MB
4. 48KB

Question 241
A process invokes the ....wrapper function to test and decrease all primitive semaphore values involved.

Question 242
The following command only displays the superblock information and not any of the block group descriptor detail information.
Select one:
1. $dumpe2fs -x /dev/sda1
2. $dumpe2fs -h /dev/sda1
3. $dumpe2fs -f /dev/sda1
4. $dumpe2fs -b /dev/sda1

Question 243
To avoid race conditions on the pipe's data structures, the kernel makes use of the .... semaphore included in the inode object.

Question 244
In a pipe_inode_info structure, ... field provides Index of first buffer containing data to be read.

Question 245
All fields in jbd2 (the journal) are written to disk in ....
Select one:
1. only unsigned form
2. little-endian order
3. only signed form
4. big-endian order

Question 246
In a pipe_inode_info structure, ... field provides Array of pipe's buffer descriptors.

Question 247
The number of Reserved GDT blocks in Ext4 Filesystem is ....
Select one:
1. 1021
2. 1024
3. 128
4. 512

Question 248
In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then the size of one blockgroup is ....
Select one:
1. 8MB
2. 32MB
3. 8GB
4. 64MB

Question 249
In a Ext2 file system, if disk block size = 1KB, # of inodes = 2048 and 512 Blockgroups, then size of node table is .... blocks.
Select one:
1. 512
2. 256
3. 1024
4. 2048

Question 250
Consider an Ext2 1.44 MB floppy disk which is initialized by make2fs with the default options - Block size: 1,024 bytes, Number of allocated inodes : 1 inode for each 8,192 bytes, Percentage of reserved blocks: 5 percent.
Compute the Number of Block group descriptors.
Select one:
1. 2
2. 30
3. 1
4. 20

Question 251
Assume a data block size of 1KB, if the file block number is 13 then compute the corresponding logical block number of the data block.
Select one:
1. i_block[11]
2. i_block[13] & first level indirectblock[3]
3. i_block[12] & first level indirectblock[1]
4. i_block[12] & first level indirectblock[2]

Question 252
In Ext2 superblock structure, ... field provides number of reserved blocks
Select one:
1. s_r_blocks_count
2. s_block_group_nr
3. s_blocks_per_group
4. s_blocks_count

Question 253
.... Allow processes to exchange information of large amounts of data
Select one:
1. Message Queues
2. Semaphore
3. Shared Memory
4. Sockets

QUESTION BANK ON VIRTUAL FILE SYSTEM, MAIN MEMORY MANAGEMENT, PROCESS ADDRESS SPACE, SYSTEMS CALS, SIGNALS

File: on-memory file system & File system calls
VIRTUAL FILE SYSTEM (CHAPTER 12)

1. Discuss Virtual File System in Linux.
2. What is the role of the Virtual Filesystem?
3. State the three main classes of File Systems supported by Linux VFS.
4. What are Disk-based filesystems? Give some examples
5. What is meant Network filesystems? Give some examples.
6. What are Special filesystems? Give some examples.
7. Categorize the following file systems into disk-based, Network and Special file system classes.
   UFS ....
   CIFS ....
   /proc ....
   NCP ....
   Reiser FS ....
   NTFS ....
   Pipe ....
   UDF ....
   Coda ....
   HFS ....
   NFS ....
   ADFS ....
   AFS ....
   AFFS ....
   XFS ....
   JFS ....
   HPFS ....
8. Describe the Common File Model used in VFS.
9. Explain the object types used in Common File Model of linux VFS.
10. Explain the following object types used in Common File Model of linux VFS
    a) Superblock object
    b) Inode object
    c) File Object
    d) Dentry object
11. With the help of a schematic, describe the interaction between processes and VFS objects by considering a simple example of three processes opening the same file, two of them using the same hard link.
12. Explain the purpose of Superblock object.
13. What information is stored in Inode Object?
14. What is a File Object?

15. Explain the purpose of dentry object.
16. How many dentry objects are created by the kernel when looking up the pathname /usr/tmp/test. List the dentry objects created.
17. Describe the four states of a dentry object.
18. Explain the following four states of a dentry object :
    a) Free b) Unused c) In Use d) Negative
19. Describe the two kinds of data structures used by linux in dentry cache to maximize efficiency in handling dentries.
20. Explain the approach used by linux to manage unused and negative dentries in dentry cache.
21. State the purpose of the following fields in a process descriptor :
    a) fs_struct fs b) files_struct files
22. A process cannot use more than .... file descriptors and kernel enforces a dynamic bound on the maximum number of file descriptors in a process to ....
23. Explain the need for a special file system.
24. What are the two numbers associated with a block device. State the purpose of each of these numbers.
25. Explain the need for File System Registration. Describe the various fields in the filesystem registration data structure file-system_type object.
26. Define mount point of a filesystem. Explain the terms fixed, any and none with respect to a mount point.
27. Write a command to mount a floppy device /dev/fp0 on a temporary directory /floppy.
28. Define namespace of a process.
29. State whether the following statement is TRUE or FALSE.
    "In most traditional Unix-like kernels, each filesystem can be mounted only once. However, in Linux - it is possible to mount the same filesystem several times." ....
30. Explain the mounting and unmounting file system operations in linux.
31. Write notes on
    a) Mounting Generic filesystems
    b) Mounting Rooted Filesystem
    c) Unmounting a Filesystem
32. Why does the kernel bother to mount the rootfs filesystem before the real one?
33. State the need for File Locking.
34. State whether the following statements are TRUE or FALSE
    Several processes may have read locks on some file region, but only one process can have a write lock on it at the same time. ....
    It is possible to get a write lock when another process owns a read lock for the same file region, and vice versa ....
35. What is meant by Advisory Locking?
36. What is meant by Mandatory Locking?
37. Distinguish between shared lock and exclusive lock. Explain why they are required.
38. Distinguish between Advisory and Mandatory Locking.

39. Explain the following two system calls that can be used to perform Advisory file locking
    a) fcntl( ) b) flock( )
40. What are the second and third arguments of fcntl() system call. Why are they required.
41. Write the system call which can be used to lock a specified region of a file.
42. Write the system call which can be used to lock only whole file.
43. Describe the purpose of fcntl() based mandatory lock named lease.
44. Describe the two steps to implement mandatory file locking in linux.
45. Describe the two ways in which a process can get or release an advisory file lock on a file.
46. State the purpose of setuid and setgid permission bits.
47. Give linux commands to perform the following operations :
    a) setuid b) setgid
48. Indicate the operations performed by the following commands :
    chmod u+s file
    chmod g+s file
    chmod 4777 file1
    chmod 2764 file2
    chmod 6764 file3
49. Write the purpose of the following linux commands:
    find / -type f -perm /6000
    find / -type f -perm /4000
    find / -type f -perm /2000
50. Explain how lease lock is implemented in linux using the system call fcntl().

MEMORY MANAGEMENT (CHAPTER 6)

1. Explain the following:
    a) Memory Management
    b) Virtual memory
    c) Protected memory
    d) Shared memory
2. Explain why the kernel must keep track of the current status of each page frame.
3. Explain the purpose of page descriptors.
4. State information of a page frame is kept in a page descriptor of type ....
5. All page descriptors are stored in the .... array.
6. Size of each page descriptor is ...... bytes
    a) 8 b) 16 c) 32 d) 64
7. Given a main memory of 4GB, determine the maximum number of page frames in this memory and the size of the mem_map array required to store all the page descriptors.
8. The .... macro yields the address of the page descriptor associated with the linear address addr.

9. The .... macro yields the address of the page descriptor associated with the page frame having number pfn.
10. State the purpose of the following macros :
    a) virt_to_page(addr) b) pfn_to_page(pfn)
11. Describe the purpose of the following fields in the page descriptor :
    a) flags b) _count c) _mapcount d) private
    e) mapping f) index g) lru
12. Explain the following :
    a) Uniform Memory Access Architecture (UMA)
    b) Non-Uniform Memory Access Architecture (NUMA)
13. Indicate the number of memory nodes in UMA architecture.
14. Indicate the maximum number of memory nodes in NUMA architectures considering both 32-bit and 64-bit architectures.
15. Maximum number of memory nodes in 32-bit architecture is .... and number of bits required in flags field to store both zone and node numbers is ....
16. Maximum number of memory nodes in 64-bit architecture is .... and number of bits required in flags field to store both zone and node numbers is ....
17. Discuss the two hardware constraints of the 80x86 32-bit architecture that may limit the way page frames can be used.
18. Explain how Linux Partitions the physical memory of every memory node into Memory zones. Describe these Memory zones
19. Which field bits in the page descriptor are used to specify the memory node number and the zone number? How many bits are used to specify memory node number and the zone number in UMA and NUMA architectures?
20. Explain the following :
    a) DMA Zone b) NORMAL Zone c) HIGH MEMORY Zone
21. Size of DMA Zone is ....MB and size of NORMAL Zone is .... MB
22. Explain what is meant by Page reclaiming. Indicate when is it required and which kernel thread is invoked to perform page reclaiming.
23. State the purpose of the kernel thread kswapd.
24. Each node has a descriptor of type .... and all node descriptors are stored in a singly linked list, whose first element is pointed to by the .... variable.
25. Explain the purpose of the following fields in memory zone descriptor :
    a) Free_pages b) pages_min c) pages_low
    d) pages_high e) lowmem_reserve
26. State the purpose of the function page_zone().
27. What are Pool of Reserved Page Frames? Why are they required? Where they are located?
28. Explain the two kernel control paths which make use of atomic memory allocation requests. State the need for having these separate memory allocation requests.
29. Write the expression for computing reserved pool size. What are the minimum and maximum values of the reserved pool size that can be stored in the variable min_free_kbytes? Also specify the directly mapped memory sizes used to compute these values.
30. Describe the approach used to build the pool of Reserved Page frames by considering the sizes of DMA Zone and NORMAL Zone.

31. State the purpose of pages_min field in the memory zone descriptor. Show how this field is used to compute values of pages_low and pages_high fields in the zone descriptor?
32. With the help of a schematic, describe the Zoned Page Frame Allocator. Explain the purpose of all the components in the system.
33. Describe the following components in the Zoned Page Frame Allocator:
    a) Zone Allocator b) Buddy System c) Per-CPU Page Frame Cache
34. Describe the SIX different functions used to request page frames allocation.
35. What is gfp_mask parameter in allocation function? Why is it required?
36. State the purpose of __GFP_DMA and __GFP_HIGHMEM flags of the gfp_mask.
37. Describe the FOUR different functions used to release page frames.
38. Describe the approach adopted on 32-bit architectures to exploit all the available RAM, up to the 64 GB supported by PAE.
39. Discuss the three different mechanisms used by the kernel to map page frames in high memory.
40. Describe the following mechanisms used by the kernel to map page frames in high memory clearly providing the approach used.
    a) Permanent Kernel Mapping
    b) Temporary Kernel mapping
    c) Non-contiguous Memory Allocation
41. What are the two major problems of Memory Management tackled by Buddy algorithm?
42. Explain the following memory management problems :
    a) External Fragmentation
    b) Internal Fragmentation
43. Describe the working of a Buddy System Algorithm through a simple example and state how this algorithm gets the name Buddy
44. What are the functions used to allocate and deallocate a block in a zone?
45. State the purpose of Per_CPU Page frame cache.
46. Describe the two per-CPU Page fame caches defined for each memory zone.
47. Explain the following Per_CPU page frame caches:
    a) Hot cache b) Cold cache
48. Write the functions used to allocate and release page frames from the Per_CPU page frame cache.
49. Describe the functions of a zone allocator
50. Explain what is meant by internal Fragmentation. Describe the approach used by Linux Kernel to reduce Internal fragmentation
51. Running a memory area allocation algorithm on top of the buddy algorithm is not particularly efficient. Discuss the better approach used in linux kernel to achieve the efficiency.
52. Explain the following :
    a) Slab b) object
53. With a suitable schematic, describe the functions of a Slab Allocator. Clearly explain the functions of each component in the slab allocator.
54. What are General and Specific caches?
55. Write the functions for allocating a slab to a cache and destroying a slab from a cache.

56. With the help of a schematic, explain the relationship between slab and object descriptors.
57. Explain what is meant by aligning objects in memory. Why is it required?
58. Explain the need for Slab Coloring mechanism.
59. Describe the approach used in slab coloring mechanism to reduce higher hardware cache misses.
60. Write the expression for computing the length of a slab clearly explaining the various parameters used in the expression.
61. With the help of a schematic, illustrate how the placement of objects inside the slab depends on the slab color.
62. Explain what is meant by General Purpose object.
63. Describe the memory Pool of a kernel component. Indicate clearly where it is located.
64. Explain the need for Non-Contiguous Memory Area.
65. Describe Non-Contiguous Memory Area Management in Linux
66. Draw the memory layout of the kernel memory (fourth GB of RAM) showing all the memory areas (NORMAL Zone (Physical Memory Mapping), Non-Contiguous Memory Area, Memory areas used for Persistent Kernel mappings and fix-mapped linear addresses), their sizes, starting and ending addresses of each of these memory areas. Also show how non-contiguous blocks are allocated in the memory area.
67. Provide the addresses stored in following parameters :
    a) PAGE_OFFSET b) PKMAP_BASE
68. State the need for inserting safety interval between the end of the physical memory mapping and the first memory area and additional safety intervals to separate non-contiguous memory areas in the kernel memory. Also indicate the size of these safety intervals.
69. What are the functions used to allocate and release Non-contiguous memory Areas.
70. Given the following Memory layout containing 10 memory areas with their sizes, determine the start and end address of each of these memory areas assuming that the start address of first memory area (MA1) is 0x70000000.

MA1	MA2	MA3	MA4	MA5	MA6	MA7	MA8	MA9	MA10
364MB	184MB	296MB	88MB	540MB	56MB	32MB	128MB	256MB	52MB

PROCESS ADDRESS SPACE (CHAPTER 7)

1. State the three functions used by the kernel to get dynamic memory.
2. What is meant by a memory region.
3. Explain what is meant by address space of a process.
4. Indicate some typical situations in which a process gets new memory regions
5. Explain the purpose of the following system calls :
   a) brk() b) execve() c) fork() d) mmap()
   e) mremap() f) munmap() g) shmat() h) shmdt()
6. What is a Page Fault Exception?
7. What are the two main causes for generating page fault exception?
8. Explain what is meant by memory descriptor. Why is it required?
9. Describe some of the important fields in the memory descriptor.
10. Explain the purpose of the following fields in the memory descriptor :
    a) mmap b) mm_rb c) mm_users d) mm_count
    e) map_count f) start_code, end_code g) start_data, end_data
    h) start_brk i) start-stack j) arg_start, arg_end k)env_start, env_end
11. Write the functions used to get a new memory descriptor and release a memory descriptor.
12. What are the two kinds of memory descriptor pointers included in every process descriptor? Why are they required?
13. Describe the memory region objects. Explain the various fields in the memory region object data structure.
14. Explain the purpose of the following fields in the memory region object :
    a) vm_mm b) vm_start c) vm_end d) vm_next e) vm_page_prot
    f) vm_flags g) vm_rb
15. Write the four methods to act on a memory region that are applicable to UMA systems.
16. The kernel finds the memory regions through the .... field of the process memory descriptor, which points to the first memory region descriptor in the list.
17. The .... field of the memory descriptor contains the number of regions owned by the process.
18. By default, a process may own up to .... different memory regions.
    a) 4096 b) 65535 c) 32,768 d) 65,536
19. With the help of a schematic, illustrate the relationships among the address space of a process, its memory descriptor, and the list of memory regions.
20. Describe the data structure used by Linux to store memory descriptors for implementing efficient search, insertion and deletion.
21. Describe the red-black tree data structure used by Linux to store memory descriptors for implementing efficient search, insertion and deletion.
22. What are the three kinds of flags associated with a page?
23. List the descriptors or entries containing three kinds of flags associated with a page.
24. State the purpose of the Read/Write, Present, and User/Supervisor flags in each Page Table entry.
25. Discuss memory region flags.

26. State the purpose of NX flag in the processor.
27. Write the low-level functions to perform the following operations on a memory region descriptor.
    a) Finding the closest region to a given address
    b) Finding a region that overlaps a given interval
    c) Finding a free interval
    d) Inserting a region in the memory descriptor list
    e) Releasing a region from the memory descriptor list
28. State the purpose of the following low-level functions in linux and provide the format of each of these functions.
    a) Find_vma() b) find_vma_intersection() c) get_unmapped_area()
    d) insert_vm_struct() e) do_mmap() f) do_munmap()
    g) vma_link() h) vma_unlink() i) split_vma()
29. Explain the purpose of do_page_fault() function.
30. What are the two input parameters required by the do_page_fault() function? How does the function distinguish between two kinds of page fault exceptions?
31. With the help of a schematic, describe the linux Page Fault Exception handler overall scheme.
32. Daw the complete flow diagram of the Page Fault Handler and explain all the steps of the Page Fault handler.
33. Explain what is meant by demand paging.
34. Explain the Copy On Write Approach in Linux
35. State the purpose of copy_mm() function.
36. Write the system call to create a lightweight process.
37. State the purpose of exit_mm() function.
38. List and explain all the APIs that can be used by the process to request and release dynamic memory.
39. Explain the following APIs that can be used by the process to request and release dynamic memory:
    a) malloc(size) b) calloc(n, size) c) realloc(ptr, size)
    d) free(addr) e) brk(addr) f) sbrk(incr)
40. State the purpose of sys_brk(addr) function.

SYSTEM CALLS & SIGNALS (CHAPTER 8 & 9)

1. What are System Calls?
2. What are APIs?
3. Distinguish between APIs and System Calls.
4. What is the relation between APIs and System Calls?
5. Draw the Control Flow Diagram of a System Call showing the relationships among the application program that invokes a system call, the corresponding wrapper routine, the system call handler, and the system call service routine.
6. What are the operations performed by a System call handler?
7. What is a signal. State the two main purposes served by a signal.
8. State the difference between regular (standard) signals and real-time signals.

9. The number of regular signals in linux is …… (numbered from … to….) and the number of POSIX real-time signals supported in linux is ………….(numbered from …. to ……).
10. The following table lists the names of the signals handled by Linux for the 80x86 architecture. Write the purpose/function associated with each of the signal and default action performed :

#	Signal Name	Purpose	Default Action
1	SIGHUP		
2	SIGINT		
3	SIGQUIT		
4	SIGILL		
5	SIGTRAP		
6	SIGABRT		
7	SIGBUS		
8	SIGFPE		
9	SIGKILL		
11	SIGSEGV		
14	SIGALRM		
15	SIGTERM		
17	SIGCHLD		
18	SIGCONT		
19	SIGSTOP		

11. Describe the following two different phases related to signal transmission/reception:
    a) Signal Generation (sending) b) Signal Delivery (Receiving)
12. What are pending signals?
13. Describe the Actions Performed by a process upon Delivering (Receiving) a Signal.
14. What are the three ways in which a process can respond to a signal?
15. List all the conditions that can generate Signals?
16. Explain what is meant by a fatal signal for a given process.
17. Distinguish between blocking and ignoring signals. Indicate the two signals which cannot be ignored, caught, or blocked, and their default actions must always be executed.
18. Explain what is meant by catching a signal.
19. With the help of a schematic, illustrate the flow of execution of the functions involved in catching a signal.

20. Explain the following system calls associated with the signals
a)kill b) tkill( ) c) tgkill( )

QUIZ QUESTIONS

Question 1
The .... stores information about the interaction between an open file and a process. This information exists only in kernel memory during the period when a process has the file open.
Select one:
1. dentry object
2. file object
3. inode object
4. superblock object

Question 2
A kernel function gets dynamic memory by invoking .... to use the slab allocator for specialized or general-purpose objects
Select one:
1. __get_free_pages( )
2. kmem_cache_alloc( ) or kmalloc( )
3. vmalloc( ) or vmalloc_32( )
4. alloc_pages

Question 3
In 80x86 architecture, in the fourth gigabyte of RAM, PKMAP_BASE refers to .... address
Select one:
1. 0x10000000
2. 0xFC000000
3. 0xC0000000
4. 0xFE000000

Question 4
The key idea behind the VFS consists of introducing a .... model capable of representing all supported filesystems.

Question 5
The .... field of the memory descriptor provides pointers to adjacent elements in the list of memory descriptors
Select one:
1. rss
2. mmlist
3. total_vm
4. hiwater_rss

Question 6
The .... function allocates page frames through the per-CPU page frame caches
Select one:
1. buffered_rmqueue( )
2. _ _free_pages_bulk( )
3. free_hot_cold_page( )
4. _ _rmqueue( )

Question 7
Inside each memory zone, page frames are handled by a component named ....
Select one:
1. Zone Allocator
2. Buddy System
3. Slab Allocator
4. Per-CPU Page frame cache

Question 8
SAY TRUE OR FALSE
"Two adjacent memory regions can be merged even if their access rights do not match."

Question 9
Maximum number of memory nodes in NUMA 32-bit architecture is ....
Select one:
1. 64
2. 1024
3. 256
4. 32

Question 10
.... field in file object provides a pointer to the Mounted filesystem containing the file.
Select one:
1. f_dentry
2. f_op
3. f_list
4. f_vfsmnt

Question 11
For each process, the kernel maintains a variable .... that points to the top of the heap.
Select one:
1. cs
2. pc
3. brk
4. sp

Question 12
SAY TRUE OR FALSE
"When the user types a command at the console, the shell process creates a new process to execute the command. As a result, a fresh address space, and thus a set of memory regions, is assigned to the new process."

Question 13
.... operation associated with inode object creates a new disk inode for a special file associated with a dentry object in some directory. The mode and rdev parameters specify, respectively, the file type and the device's major and minor numbers.
Select one:
1. mknod(dir, dentry, mode, rdev)
2. follow_link(inode, nameidata)
3. lookup(dir, dentry, nameidata)
4. link(old_dentry, dir, new_dentry)

Question 14
The ext2_truncate( ) function essentially scans the disk inode's i_block array to locate all data blocks and all blocks used for the indirect addressing. These blocks are then released by repeatedly invoking ....
Select one:
1. ext2_alloc_block( )
2. ext2_truncate( )
3. ext2_free_blocks( )
4. ex2_get_block()

Question 15
Size of each page descriptor is ...... bytes
Select one:
1. 8
2. 16
3. 64
4. 32

Question 16
To get better system performance, ... is used to quickly satisfy the allocation requests for single page frames.
Select one:
1. Slab Allocator
2. Per-CPU Page frame cache
3. Zone Allocator
4. Buddy System

Question 17
A problem called .... is caused by a mismatch between the size of the memory request and the size of the memory area allocated to satisfy the request.
Select one:
1. external fragmentation
2. internal fragmentation
3. non-contiguous memory allocation
4. contiguous memory allocation

Question 18
The .... field in the process descriptor points to the memory descriptor used by the process when it is in execution.
Select one:
1. mm
2. mmdrop( )
3. active_mm
4. mmput( )

Question 19
A signal is .... for a given process if delivering the signal causes the kernel to kill the process.
Select one:
1. fatal
2. unblocked
3. ignored
4. blocked

Question 20
The .... field of the memory descriptor provides size of the process address space (number of pages).
Select one:
1. hiwater_rss
2. total_vm
3. rss
4. mmlist

Question 21
In NUMA 32-bit architecture, the number of bits required in flags field to store both zone and node numbers is ....
Select one:
1. 12
2. 3
3. 10
4. 8

Question 22

The .... utility program is invoked by the boot script to check the value stored in s_mount_state field of the superblock on disk; if it is not equal to EXT2_VALID_FS, the filesystem was not properly unmounted, and therefore .... starts checking all disk data structures of the filesystem.
(WRITE ONE ANSWER FOR BOTH)

Question 23

A kernel function gets dynamic memory by invoking .... to get pages from the zoned page frame allocator
Select one:
1. vmalloc( )
2. __get_free_pages( )
3. vmalloc_32( )
4. kmem_cache_alloc( ) or kmalloc( )

Question 24

In NUMA 64-bit architecture, the number of bits required in flags field to store both zone and node numbers is ....
Select one:
1. 10
2. 8
3. 3
4. 12

Question 25

When the slab allocator creates a new slab, it relies on the zoned page frame allocator to obtain a group of free contiguous page frames by invoking the .... function
Select one:
1. kmem_getpages( )
2. slab_destroy( )
3. cache_grow( )
4. kmem_freepages( )

Question 26

All information related to the process address space is included in an object called the memory descriptor of type ....
Select one:
1. mem_map
2. page_dat_t
3. mm_struct
4. vm_area_struct

Question 27
.... operation associated with superblock object removes every reference to the inode from the VFS data structures and, if the inode no longer appears in any directory, invokes the delete_inode superblock method to delete the inode from the filesystem.
Select one:
1. dirty_inode(inode)
2. generic_drop_inode( )
3. sync_fs(sb, wait)
4. put_inode(inode)

Question 28
SAY TRUE OR FALSE
"The pages_min field of the zone descriptor stores the number of reserved page frames inside the zone. This field plays also a role for the page frame reclaiming algorithm, together with the pages_low and pages_high fields. The pages_low field is always set to 5/4 of the value of pages_min, and pages_high is always set to 3/2 of the value of pages_min."

Question 29
Categorize the XFS file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 30
The .... function is invoked to get a new memory descriptor.
Select one:
1. mm_users
2. kmem_cache_alloc( )
3. mm_alloc( )
4. mm_count

Question 31
The mm_alloc( ) calls .... function in turn to store and initialize the new memory descriptor in a slab allocator cache.
Select one:
1. kmem_cache_alloc( )
2. mm_users
3. mm_count
4. mm_alloc( )

Question 32
The .... function releases page frames to the per-CPU page frame caches
Select one:
1. free_hot_cold_page( )
2. buffered_rmqueue( )
3. _ _free_pages_bulk( )
4. _ _rmqueue( )

Question 33
.... system call allows a process to lock a whole file and not a region of the file.

Question 34
On NUMA 32-bit architectures, flags field in the page descriptor has .... bits for the zone index and .... bits for the node number.
Select one:
1. 2, 6
2. 2, 10
3. 2, 1
4. 2,4

Question 35
Specific caches are created by the .... function
Select one:
1. kmem_cache_shrink( )
2. kmem_cache_create( )
3. kmem_freepages( )
4. kmem_cache_init( )

Question 36
An .... signal is always delivered, and there is no further action.
Select one:
1. ignored
2. fatal
3. blocked
4. unblocked

Question 37
Given a main memory of 4GB, determine the size of the mem_map array required to store all the page descriptors in memory.
Select one:
1. 32MB
2. 32KB
3. 4MB
4. 32GB

Question 38
In 80x86 architecture, in the fourth gigabyte of RAM, PAGE_OFFSET refers to .... address
Select one:
1. 0xFC000000
2. 0x10000000
3. 0xC0000000
4. 0xFE000000

Question 39
Given a memory area MA2 with start address 0xD0000000 and size 512 MB, compute the end address
Select one:
1. 0xEFFFFFFF
2. 0xBFFFFFFF
3. 0xCFFFFFFF
4. 0xFFFFFFFF

Question 40
Categorize the UDF file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 41
.... field in the zone descriptor Specifies how many page frames in each zone must be reserved for handling low-on-memory critical situations.
Select one:
1. pages_min
2. lowmem_reserve
3. pages_low
4. pages_high

Question 42
Given a main memory of 8GB, determine the maximum number of page frames in this memory considering Linux default page frame size.
Select one:
1. 2 Giga page frames
2. 2Mega page frames
3. 4 Kilo page frames
4. 2 Kilo page frames

Question 43
Categorize the Pipe file system (pipefs) into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 44
Given a main memory of 8GB, determine the size of the mem_map array required to store all the page descriptors in memory.
Select one:
1. 128MB
2. 64KB
3. 64GB
4. 64MB

Question 45
Kernel uses .... function to destroy a temporary kernel mapping
Select one:
1. kunmap_atomic()
2. kunmap( )
3. kmap( )
4. kmap_atomic()

Question 46
By default, a process may own up to .... different memory regions.
Select one:
1. 65535
2. 32768
3. 65536
4. 4096

Question 47
Maximum number of memory nodes in NUMA 64-bit architecture is ....
Select one:
1. 256
2. 1024
3. 64
4. 32

Question 48
.... state of dentry object indicates that the dentry object is not currently used by the kernel. The d_count usage counter of the object is 0, but the d_inode field still points to the associated inode. The dentry object contains valid information, but its contents may be discarded if necessary in order to reclaim memory.

Question 49
In Linux, the function used for finding the closest region to a given address is ....
Select one:
1. find_vma( )
2. insert_vm_struct()
3. get_unmapped_area( )
4. find_vma_intersection( )

Question 50
The .... function destroys all the slabs in a cache by invoking slab_destroy( ) iteratively
Select one:
1. kmem_cache_create( )
2. kmem_cache_shrink( )
3. kmem_cache_init( )
4. kmem_freepages( )

Question 51
Categorize the JFS file system into .... class.
Select one:
1. Network file system
2. Disk-based file system
3. Special file system
4. None of these classes

Question 52
The .... function changes the size of a memory area previously allocated by malloc() or calloc( ).
Select one:
1. calloc(n,size)
2. brk( )
3. realloc(ptr,size)
4. sys_brk(addr)

Question 53
SAY TRUE OR FALSE
"Like a semaphore used to protect a critical region in code, the lock is considered "advisory" because it doesn't work unless other processes cooperate in checking the existence of a lock before accessing the file. "
Question 54
Categorize the NTFS file system into .... class.
Select one:
1. Network file system
2. Special file system
3. None of these classes
4. Disk-based file system

Question 55
Categorize the NCP file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 56
.... macro is used to request 2^order contiguous page frames. It returns the address of the descriptor of the first allocated page frame or returns NULL if the allocation failed.
Select one:
1. alloc_pages(gfp_mask, order)
2. __get_free_page(gfp_mask)
3. alloc_page(gfp_mask)
4. __get_free_pages(gfp_mask, order)

Question 57
In Linux, the function used for Finding a free interval is ....
Select one:
1. get_unmapped_area( )
2. insert_vm_struct()
3. find_vma( )
4. find_vma_intersection( )

Question 58
Given a memory area MA1 with start address 0x80000000 and size 1024 MB, compute the end address
Select one:
1. 0xEFFFFFFF
2. 0xEFFFFFFF
3. 0xCFFFFFFF
4. 0xBFFFFFFF

Question 59
Because processes are created and destroyed quite frequently, the kernel wastes time allocating and deallocating the page frames containing the same memory areas repeatedly; the .... allows these to be saved in a cache and reused quickly thereby improving the performance.
Select one:
1. Buddy System
2. Slab Allocator
3. Zone Allocator
4. Per-CPU Page frame cache

Question 60
In 32-bit architetcure, the .... zone includes page frames that cannot be directly accessed by the kernel through the linear mapping in the fourth gigabyte of linear address space.
Select one:
1. ZONE_NORMAL
2. ZONE_DMA
3. ZONE_HIGHMEM
4. ZONE_LOWMEM

Question 61
The .... field of the memory region object or descriptor provides first linear address inside the memory region.
Select one:
1. vm_next
2. vm_start
3. vm_end
4. vm_mm

Question 62
SAY TRUE OR FALSE
"A single API function could make several system calls."

Question 63
SAY TRUE OR FALSE
"A process may create an IPC-shared memory region to share data with other cooperating processes. In this case, the kernel assigns a new memory region to the process to implement this construct."

Question 64
The .... field of the memory descriptor provides number of memory regions in the process address space.
Select one:
1. mm_rb
2. mmap_cache
3. map_count
4. mmap

Question 65
The .... field of the memory descriptor provides maximum number of page frames ever owned by the process.
Select one:
1. rss
2. mmlist
3. total_vm
4. hiwater_rss

Question 66
.... operation associated with superblock object is Invoked when flushing the filesystem to update filesystem-specific data structures on disk (used by journaling filesystems).
Select one:
1. generic_drop_inode( )
2. put_inode(inode)
3. sync_fs(sb, wait)
4. dirty_inode(inode)

Question 67
Categorize the UFS file system into .... class.
Select one:
1. Special file system
2. Disk-based file system
3. Network file system
4. None of these classes

Question 68
.... operation associated with file object flushes the file by writing all cached data to disk.
Select one:
1. ioctl(inode, file, cmd, arg)
2. poll(file, poll_table)
3. fsync(file, dentry, flag)
4. mmap(file, vma)

Question 69
Kernel uses .... function to destroy a permanent kernel mapping
Select one:
1. kmap()
2. kunmap( )
3. kmap_atomic()
4. kunmap_atomic

Question 70
.... field in superblock object provides Pointer to superblock information of a specific filesystem.
Select one:
1. s_id
2. s_fs_info
3. s_type
4. s_op

Question 71
……………………… operation associated with file object sends a command to an underlying hardware device. This method applies only to device files.
Select one:
1. fsync(file, dentry, flag)
2. poll(file, poll_table)
3. ioctl(inode, file, cmd, arg)
4. mmap(file, vma)

Question 72
In Linux, the function used for Allocating a Linear Address Interval is ………………
Select one:
1. vma_unlink( )
2. vma_link( )
3. do_munmap( )
4. do_mmap( )

Question 73
A process cannot use more than ………….. file descriptors and kernel enforces a dynamic bound on the maximum number of file descriptors in a process to ………..
Select one:
1. 1048576, 1024
2. 65536, 1024
3. 1048576, 512
4. 65536, 512

Question 74
In the reverse operation, page frames assigned to a slab can be released by invoking the ………………….. function.
Select one:
1. cache_grow( )
2. kmem_freepages( )
3. kmem_getpages( )
4. slab_destroy( )

Question 75
………….. field in superblock object provides pointer to Filesystem type structure
Select one:
1. s_fs_info
2. s_id
3. s_op
4. s_type

Question 76
If .... of 3-bit error_code parameter of do_page_fault() function is set, the exception was caused by an invalid access right.
Select one:
1. bit 0
2. bit 1
3. bit 2
4. bit 3

Question 77
The function insert_vm_struct( ) in turn invokes the .... function for Inserting a memory region in the linked list and the red-black tree of the memory descriptor.
Select one:
1. vma_unlink( )
2. vma_link( )
3. do_munmap( )
4. do_mmap( )

Question 78
The directory on which a filesystem is mounted is called the ....

Question 79
The .... field in the process descriptor points to the memory descriptor owned by the process.
Select one:
1. mm
2. mmdrop( )
3. mmput( )
4. active_mm

Question 80
.... field in inode object provides pointer to superblock object

Question 81
The slab allocator takes advantage of the free unused bytes to .... the slab which is used simply to subdivide the slabs and allow the memory allocator to spread objects out among different linear addresses. In this way, the kernel obtains the best possible performance from the microprocessor's hardware cache.

Question 82
.... state of dentry object indicates that the dentry object contains no valid information and is not used by the VFS. The corresponding memory area is handled by the slab allocator.

Question 83
A signal is not delivered as long as it is ....
Select one:
1. fatal
2. blocked
3. unblocked
4. ignored

Question 84
When a process in User Mode invokes the brk( ) system call, the kernel executes the .... function.
Select one:
1. calloc(n,size)
2. sys_brk(addr)
3. sbrk(incr)
4. realloc(ptr,size)

Question 85
SAY TRUE OR FALSE
"Page fault exceptions are also caused by programming errors such as access rights violation"

Question 86
SAY TRUE OR FALSE
"To store the memory regions of a process, Linux uses both a linked list and a red-black tree. Both data structures contain pointers to the same memory region descriptors."

Question 87
SAY TRUE OR FALSE
"If the page does not have any access rights, the Present bit is cleared so that each access generates a Page Fault exception. "

Question 88
Categorize the /dev into .... class.
Select one:
1. Network file system
2. Disk-based file system
3. Special file system
4. Not a file system

Question 89
SAY TRUE OR FALSE
"If the page has both Write and Share access rights, the Read/Write bit is reset."

Question 90
.... operation associated with superblock object is Invoked when the inode is marked as modified and used by filesystems such as ReiserFS and Ext3/4 to update the filesystem journal on disk.
Select one:
1. generic_drop_inode( )
2. put_inode(inode)
3. sync_fs(sb, wait)
4. dirty_inode(inode)

Question 91
The .... macro yields the address of the page descriptor associated with the page frame having number pfn.
Select one:
1. pfn_to_page(pfn)
2. page_to_pfn(addr)
3. virt_to_page(addr)
4. page_to_virt(pfn)

Question 92
Linux's kernel memory allocator uses Slab Allocator algorithm on top of a .... approach
Select one:
1. Power-of-two free lists
2. Buddy system
3. Dynix allocator
4. McKusick-Karels allocator

Question 93
The .... field of the memory descriptor provides a Pointer to the last referenced memory region object.
Select one:
1. mm_rb
2. mmap_cache
3. map_count
4. mmap

Question 94
The .... stores information concerning a mounted filesystem. For disk-based filesystems, this object usually corresponds to a filesystem control block stored on disk.
Select one:
1. superblock object
2. file object
3. inode object
4. dentry object

Question 95
The technique adopted by Linux to solve the external fragmentation problem is based on the well-known .... algorithm.
Select one:
1. Power-of-two free lists
2. Dynix allocator
3. Buddy system
4. McKusick-Karels allocator

Question 96
The size of a DMA zone is ....
Select one:
1. 16MB
2. 896MB
3. 128MB
4. 256MB

Question 97
State whether the following statements are TRUE or FALSE.
“Several processes may have read locks on some file region, but only one process can have a write lock on it at the same time.” ....

Question 98
The .... function releases an object previously allocated by the slab allocator to some kernel function.
Select one:
1. kmem_cache_free( )
2. vmalloc( )
3. vfree( )
4. kmem_cache_alloc( )

Question 99
.... field in dentry object provides pointer to Inode associated with filename.
Select one:
1. d_sb
2. d_lru
3. d_op
4. d_inode

Question 100
.... field in file object provides dentry object associated with the file.
Select one:
1. f_vfsmnt
2. f_dentry
3. f_list
4. f_op

Question 101
SAY TRUE OR FALSE
"Each memory region descriptor identifies a linear address interval."

Question 102
The slab allocator assigns a new slab to the cache by invoking the .... function
Select one:
1. cache_grow( )
2. kmem_getpages( )
3. slab_destroy( )
4. kmem_freepages( )

Question 103
Categorize the NFS file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. None of these classes
4. Special file system

Question 104
The .... function decreases mm_count field of memory descriptor and, if it becomes zero, releases the mm_struct data structure, i.e., memory descriptor.
Select one:
1. active_mm
2. mmdrop( )
3. mm
4. mmput( )

Question 105
Categorize the HFS file system into .... class.
Select one:
1. Network file system
2. None of these classes
3. Disk-based file system
4. Special file system

Question 106
.... in each memory zone stores page frames whose contents are likely to be included in the CPU's hardware cache.
Select one:
1. A cold cache
2. Unified Cache
3. Split Cache
4. A hot cache

Question 107
.... in each memory zone stores page frames for DMA operation.
Select one:
1. Split Cache
2. Unified Cache
3. A hot cache
4. A cold cache

Question 108
The .... function is invoked to destroy a slab and release the corresponding page frames to the zoned page frame allocator.
Select one:
1. cache_grow( )
2. kmem_freepages( )
3. kmem_getpages( )
4. slab_destroy( )

Question 109
The .... function requests an array consisting of n elements of size size; if the allocation succeeds, it initializes the array components to 0 and returns the linear address of the first element.
Select one:
1. sys_brk(addr)
2. realloc(ptr,size)
3. brk( )
4. calloc(n,size)

Question 110
.... field in file object provides Pointer to file operation table
Select one:
1. f_op
2. f_vfsmnt
3. f_list
4. f_dentry

Question 111
" To make use of mandatory locks, mandatory locking must be enabled both on the filesystem that contains the file to be locked, and on the file itself. "
SAY TRUE OR FALSE

Question 112
To mount an Ext3 filesystem stored in the /dev/sda1 partition on the / mount point with the ordered mode, the system administrator can type the command:

mount -t ext3 -o data=......... /dev/sda2 /

Question 113
.... field in inode object provides pointer to default file operations

Question 114
The .... function implements the buddy system strategy for freeing page frames.
Select one:
1. _ _rmqueue( )
2. _ _free_pages_bulk( )
3. free_hot_cold_page( )
4. buffered_rmqueue( )

Question 115
The .... field of the memory descriptor stores the number of lightweight processes that share the mm_struct data structure.
Select one:
1. kmem_cache_alloc( )
2. mm_alloc( )
3. mm_count
4. mm_users

Question 116
.... field in superblock object provides Name of the block device containing the superblock
Select one:
1. s_id
2. s_type
3. s_fs_info
4. s_op

Question 117
.... field in file object provides Pointers for generic file object list.
Select one:
1. f_vfsmnt
2. f_op
3. f_list
4. f_dentry

Question 118
A kernel function gets dynamic memory by invoking .... to get a noncontiguous memory area.
Select one:
1. vmalloc( ) or vmalloc_32( )
2. kmem_cache_alloc( )
3. __get_free_pages( )
4. kmalloc( )

Question 119
The .... function modifies the size of the heap directly; the addr parameter specifies the new value of current->mm->brk, and the return value is the new ending address of the memory region (the process must check whether it coincides with the requested addr value).
Select one:
1. realloc(ptr,size)
2. sys_brk(addr)
3. brk(addr)
4. calloc(n,size)

Question 120
In Linux, the function used for Inserting a region in the memory descriptor list is ....
Select one:
1. find_vma_intersection( )
2. find_vma( )
3. get_unmapped_area( )
4. insert_vm_struct()

Question 121
SAY TRUE OR FALSE
"Every red-black tree with n internal nodes has a height of at most 2 x log(n + 1)."

Question 122
SAY TRUE OR FALSE
"Objects that have the same offset within different slabs will, with a relatively high probability, end up mapped in the same cache line. The cache hardware might therefore waste memory cycles transferring two objects from the same cache line back and forth to different RAM locations, while other cache lines go underutilized."

Question 123
The .... function is invoked during system initialization to set up the general caches.
Select one:
1. kmem_freepages( )
2. kmem_cache_shrink( )
3. kmem_cache_init( )
4. kmem_cache_create( )

Question 124
The kernel subsystem that handles the memory allocation requests for groups of contiguous page frames is called the ....
Select one:
1. Zoned page frame allocator
2. Buddy System
3. Per-CPU Page frame cache
4. Slab Allocator

Question 125
.... field in dentry object provides pointers for the list of unused dentries
Select one:
1. d_inode
2. d_sb
3. d_op
4. d_lru

Question 126
SAY TRUE OR FALSE
"While the system boots, the kernel finds the major number of the disk that contains the root filesystem in the ROOT_DEV variable."

Question 127
.... operation associated with inode object translates a symbolic link specified by an inode object; if the symbolic link is a relative pathname, the lookup operation starts from the directory specified in the second parameter.
Select one:
1. lookup(dir, dentry, nameidata)
2. follow_link(inode, nameidata)
3. mknod(dir, dentry, mode, rdev)
4. link(old_dentry, dir, new_dentry)

Question 128
The head of the red-black tree is referenced by the .... field of the memory descriptor.

Question 129
.....................is a short message sent to a process by the kernel to notify the process that certain event has occurred.
Select one:
1. Kernel Control path
2. System call
3. API
4. Signal

Question 130
.... operation associated with superblock object is Invoked when the inode is released its reference counter is decreased to perform filesystem-specific operations.
Select one:
1. put_inode(inode)
2. sync_fs(sb, wait)
3. dirty_inode(inode)
4. generic_drop_inode( )

Question 131
Linux implements a memory region by means of an object of type ....
Select one:
1. mm_struct
2. mem_map
3. page_dat_t
4. vm_area_struct

Question 132
When a process deletes a file or truncates it to 0 length, all its data blocks must be reclaimed. This is done by ...., which receives the address of the file's inode object as its parameter.
Select one:
1. ext2_truncate( )
2. ext2_free_blocks( )
3. ex2_get_block()
4. ext2_alloc_block( )

Question 133
If the amount of physical memory that is directly mapped in the kernel's fourth gigabyte of linear addresses is 1024MB, then the reserved pool size used for atomic memory allocation requests to be used only on low-on memory conditions is ....
Select one:
1. 32684KB
2. 65536KB
3. 128KB
4. 1024KB

Question 134
When the kernel must delete a linear address interval from the address space of the current process, it uses the .... function.
Select one:
1. do_munmap( )
2. vma_unlink( )
3. vma_link( )
4. do_mmap( )

Question 135
.... field in superblock object provides Pointers for a list of superblock objects of a given filesystem type
Select one:
1. s_type
2. s_id
3. s_instances
4. s_fs_info

Question 136
Categorize the Coda file system into .... class.
Select one:
1. Disk-based file system
2. None of these classes
3. Network file system
4. Special file system

Question 137
In .... journaling mode, only changes to filesystem metadata are logged to the journal and is the fastest mode.

Question 138
The root of the RB tree must be ....
Select one:
1. black
2. yellow
3. red
4. blue

Question 139
Categorize the ADFS file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 140
The .... field of the memory descriptor provides number of page frames allocated to the Process.
Select one:
1. rss
2. total_vm
3. hiwater_rss
4. mmlist

Question 141
The .... stores general information about a specific file. For disk-based filesystems, this object usually corresponds to a file control block stored on disk.
Select one:
1. superblock object
2. file object
3. dentry object
4. inode object

Question 142
" Mark the files as candidates for mandatory locking by setting their set-group bit (SGID) and clearing the group-execute permission bit."
SAY TRUE OR FALSE.

Question 143
The kernel represents intervals of linear addresses by means of resources called ...., which are characterized by an initial linear address, a length, and some access rights.

Question 144
.... is used to store all information related to the process address space.
Select one:
1. Process Descriptor
2. File descriptor or inode
3. Page Descriptor
4. Memory Descriptor

Question 145
State whether the following statements are TRUE or FALSE.
"It is possible to get a write lock when another process owns a read lock for the same file region, and vice versa" ....

Question 146
.... operation associated with file object performs a memory mapping of the file into a process address space
Select one:
1. mmap(file, vma)
2. ioctl(inode, file, cmd, arg)
3. fsync(file, dentry, flag)
4. poll(file, poll_table)

Question 147
Given a memory area MA1 with start address 0xF0000000 and size 256 MB, compute the end address
Select one:
1. 0xEFFFFFFF
2. 0xFFFFFFFF
3. 0xBFFFFFFF
4. 0Xcfffffff

Question 148
.... field in the node descriptor provides pointer to array of page descriptors of the node.
Select one:
1. node_mem_map
2. node_start_pfn
3. node_zonelists
4. node_zones

Question 149
The .... is an abstraction layer between the application program and the filesystem implementations

Question 150
The .... function allocates a noncontiguous memory area to the kernel.
Select one:
1. vmalloc( )
2. vfree( )
3. kmem_cache_alloc( )
4. kmem_cache_free( )

Question 151
In Linux, the function used for Finding a region that overlaps a given interval is ....
Select one:
1. insert_vm_struct()
2. get_unmapped_area( )
3. find_vma_intersection( )
4. find_vma( )

Question 152
Kernel uses .... function to establish a temporary kernel mapping.
Select one:
1. kmap_atomic()
2. kmap()
3. kunmap( )
4. kunmap_atomic()

Question 153
............................is a function definition that specifies how to obtain a given service.
Select one:
1. Application Programmer Interface (API)
2. Re-entrant
3. Signal
4. System call

Question 154
.... system call destroys a memory mapping for a file, thus contracting the process address space.
Select one:
1. brk()
2. mmap_cache
3. munmap()
4. mmap()

Question 155
SAY TRUE OR FALSE
"A permanent kernel mapping can be established in interrupt handlers and deferrable functions."

Question 156
The .... field of the memory descriptor provides a Pointer to the head of the list of memory region objects.
Select one:
1. mm_rb
2. mmap
3. mmap_cache
4. map_count

Question 157
Given a memory area MA1 with start address 0xC0000000 and size 256 MB, compute the end address
Select one:
1. 0xBFFFFFFF
2. 0xFFFFFFFF
3. 0xEFFFFFFF
4. 0xCFFFFFFF

Question 158
Ctrl & Z key combinations generate .... signal.
Select one:
1. SIGKILL
2. SIGQUIT
3. SIGINT
4. SIGSTP

Question 159
.... operation associated with inode object searches a directory for an inode corresponding to the filename included in a dentry object.
Select one:
1. follow_link(inode, nameidata)
2. link(old_dentry, dir, new_dentry)
3. lookup(dir, dentry, nameidata)
4. mknod(dir, dentry, mode, rdev)

Question 160
Mount the filesystem where mandatory locking is required using the .... option in the .... command.
Select one:
1. -o writeback, umount
2. -o mand, mount
3. -o writeback, mount
4. -o mand, umount

Question 161
The ……………………. field of the memory region object or descriptor provides a Pointer to the memory descriptor that owns that memory region.
Select one:
1. vm_start
2. vm_next
3. vm_mm
4. vm_end

Question 162
Categorize the HPFS file system into ……………………….. class.
Select one:
1. Special file system
2. None of these classes
3. Disk-based file system
4. Network file system

Question 163
If …………. of 3-bit error_code parameter of do_page_fault() function is clear, the exception was caused by an access to a page that is not present (the Present flag in the Page Table entry is clear).
Select one:
1. bit 0
2. bit 1
3. bit 2
4. bit 3

Question 164
Ctrl & C key combinations generate ………………… signal
Select one:
1. SIGQUIT
2. SIGKILL
3. SIGINT
4. SIGSTP

Question 165
The ………………… function removes a memory region from the linked list and the red-black tree of the memory descriptor.
Select one:
1. do_munmap( )
2. split_vma( )
3. vma_unlink( )
4. vma_link( )

Question 166
SAY TRUE OR FALSE
"At any time, only one pending signal of a given type may exist for a process; additional pending signals of the same type to the same process are not queued but simply discarded."

Question 167
Each filesystem consists of two kinds of blocks: those containing the so-called .... and those containing regular data.

Question 168
.... state of dentry object indicates that the inode associated with the dentry does not exist because the corresponding disk inode has been deleted and the d_inode field of the dentry object is set to NULL.

Question 169
The .... of a process consists of all linear addresses that the process is allowed to use.

Question 170
The .... function is used in Buddy system to find a free block in a zone.
Select one:
1. _ _rmqueue( )
2. _ _free_pages_bulk( )
3. free_hot_cold_page( )
4. buffered_rmqueue( )

Question 171
------------- system call creates a memory mapping for a file, thus enlarging the process address space.
Select one:
1. munmap()
2. brk()
3. mmap()
4. mmap_cache

Question 172
The .... function releases noncontiguous memory areas created by vmalloc( )
Select one:
1. vmalloc( )
2. kmem_cache_free( )
3. kmem_cache_alloc( )
4. vfree( )

Question 173
To speed up searching, insertion and deletion operations, Linux stores memory region descriptors in data structures called ....
Select one:
1. hash table
2. red-black trees
3. singly linked list
4. doubly linked list
Question 174
The dentry hash table is implemented by means of a dentry_hashtable array in Linux. The array's size usually depends on the amount of RAM installed in the system; the default value is .... entries per megabyte of RAM.
Select one:
1. 64
2. 256
3. 128
4. 512
Question 175
Frequent requests and releases of groups of contiguous page frames of different sizes may lead to a situation in which several small blocks of free page frames are "scattered" inside blocks of allocated page frames. As a result, it may become impossible to allocate a large block of contiguous page frames, even if there are enough free pages to satisfy the request. this is a well-known memory management problem called ....
Select one:
1. non-contiguous memory allocation
2. contiguous memory allocation
3. internal fragmentation
4. external fragmentation
Question 176
The .... macro yields the address of the page descriptor associated with the linear address addr.
Select one:
1. page_to_virt(pfn)
2. page_to_pfn(addr)
3. pfn_to_page(pfn)
4. virt_to_page(addr)
Question 177
Identify all the meta data in an Ext2 file system :
Select one or more:
1. inode bitmap blocks
2. blocks used for indirect addressing (indirection blocks)
3. data bitmap blocks
4. superblocks
5. free blocks
6. inodes
7. group block descriptors
8. data blocks

Question 178
The component of Zoned Page frame allocator named .... receives the requests for allocation and deallocation of dynamic memory. In the case of allocation requests, the component searches a memory zone that includes a group of contiguous page frames that can satisfy the request.
Select one:
1. Zone Allocator
2. Buddy System
3. Slab Allocator
4. Per-CPU Page frame cache

Question 179
The area of main memory that contains a cache is divided into .... each of which consists of one or more contiguous page frames that contain both allocated and free objects.

Question 180
The .... function differs from the other heap management functions because it is the only one implemented as a system call.
Select one:
1. brk( )
2. realloc(ptr,size)
3. sys_brk(addr)
4. calloc(n,size)

Question 181
SGI's ...... and IBM's ...., limit themselves to logging the operations affecting metadata.
Select one:
1. VFS, XFS
2. JFS, XFS
3. XFS,JFS
4. JFS, VFS

Question 182
In 32-bit architecture, ... allow the kernel to establish long-lasting mappings of high-memory page frames into the kernel address spae using a dedicated Page Table in the master kernel page tables. But this mapping may block the current process when no free Page Table entries exist that can be used as "windows" on the page frames in high memory.
Select one:
1. Permanent kernel mappings
2. Non-Contiguous Memory Allocation
3. Contiguous Memory Allocation
4. Temporary kernel mappings

Question 183
Categorize the AFFS file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 184
.... system call can be used to lock an arbitrary region of a file (even a single byte) or to lock the whole file (including data appended in the future).

Question 185
On UMA 64-bit architectures, flags field in the page descriptor has .... bits for the zone index and .... bits for the node number.
Select one:
1. 2, 10
2. 2, 6
3. 2,4
4. 2, 1

Question 186
The .... stores information about the linking of a directory entry (that is, a particular name of the file) with the corresponding file. Each disk-based filesystem stores this information in its own particular way on disk.
Select one:
1. file object
2. superblock object
3. dentry object
4. inode object

Question 187
.... field in inode object provides Number of hard links

Question 188
On NUMA 64-bit architectures, flags field in the page descriptor has .... bits for the zone index and .... bits for the node number.
Select one:
1. 2, 6
2. 2, 1
3. 2,4
4. 2, 10

Question 189
The .... field of the memory descriptor provides a Pointer to the root of the red-black tree of memory region objects.
Select one:
1. mm_rb
2. mmap_cache
3. map_count
4. mmap

Question 190
Given a memory area MA with start address 0x70000000 and size 364 MB, compute the end address
Select one:
1. 0x85FFFFFF
2. 0x867FFFFF
3. 0x86BFFFFF
4. 0x83FFFFFF

Question 191
SAY TRUE OR FALSE
"In a traditional Unix system, there is only one tree of mounted filesystems: starting from the system's root filesystem, each process can potentially access every file in a mounted filesystem by specifying the proper pathname.
In this respect, Linux is more refined: every process might have its own tree of mounted filesystems the so-called namespace of the process."

Question 192
.... operation is used as second argument of flock system call to apply write lock on the file.
Select one:
1. SET_LCK
2. LOCK_SH
3. LOCK_UN
4. LOCK_EX

Question 193
In Linux the system calls must be invoked by executing the .... Assembly instruction.
Select one:
1. int $0x180
2. int $0x80
3. int $0x128
4. int $0x90

Question 194
Categorize the /proc file system into .... class.
Select one:
1. None of these classes
2. Special file system
3. Network file system
4. Disk-based file system

Question 195
The following command is used to set the setgid and reset the group-execute permission bit of the file myscript.
Select one:
1. chmod 4777 myscript
2. $ chmod 2777 myscript
3. $ chmod 2764 myscript
4. chmod 4764 myscript

Question 196
The size of a NORMAL zone is ....
Select one:
1. 256MB
2. 128MB
3. 896MB
4. 16MB

Question 197
Linux stores memory descriptors in data structures called .... for efficient insertion, deletion & search operations.
Select one:
1. Hash Table
2. Doubly linked list
3. Red-black trees
4. Singly linked list

Question 198
State information of a page frame is kept in a page descriptor of type ....

Question 199
SAY TRUE OR FALSE
"vm_end - vm_start denotes the length of the memory region."

Question 200
.... field in inode object provides pointer to inode operations

Question 201
………………. field in dentry object provides pointer to Dentry methods.
Select one:
1. d_op
2. d_lru
3. d_inode
4. d_sb

Question 202
……………….. system call destroys a memory mapping for a file, thus contracting the process address space.
Select one:
1. brk()
2. munmap()
3. mmap_cache
4. mmap()

Question 203
All memory descriptors are stored in a …………………..
Select one:
1. hash table
2. singly linked list
3. Binary tree
4. doubly linked list

Question 204
The ………………….. field of the memory region object or descriptor provides First linear address after that memory region.
Select one:
1. vm_mm
2. vm_next
3. vm_end
4. vm_start

Question 205
Signals that have been generated but not yet delivered are called ……………… signals .

Question 206
Categorize the CIFS file system into ……………………….. class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 207
Kernel uses .... function to establish a permanent kernel mapping.
Select one:
1. kunmap_atomic
2. kmap()
3. kmap_atomic()
4. kunmap( )

Question 208
.... field in superblock object provides pointer to Superblock methods structure.
Select one:
1. s_id
2. s_fs_info
3. s_type
4. s_op

Question 209
How many dentry objects are created by the kernel when looking up the pathname /usr/tmp/test.
Select one:
1. 2
2. 4
3. 1
4. 3

Question 210
Categorize the AFS file system into .... class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 211
All page descriptors are stored in the .... array in dynamic memory.

Question 212
Each node has a descriptor of type .... and all node descriptors are stored in a singly linked list, whose first element is pointed to by the .... variable.
Select one:
1. node_mem_map, node_start_pfn
2. pg_data_t, node_start_pfn
3. node_mem_map, pgdat_list
4. pg_data_t, pgdat_list

Question 213
The .... field is the main usage counter of the memory descriptor where all "users" in mm_users count as one unit.
Select one:
1. mm_count
2. kmem_cache_alloc( )
3. mm_users
4. mm_alloc( )

Question 214
The children of a red node in RB tree must be ....
Select one:
1. yellow
2. black
3. red
4. blue

Question 215
.... system call changes the heap size of the process.
Select one:
1. mmap_cache
2. mmap_cache
3. brk()
4. munmap()

Question 216
SAY TRUE OR FALSE
"If the NX bit is supported and the page does not have the Execute access right, then NX bit is set."

Question 217
.... field in dentry object provides pointer to Superblock object of the file.
Select one:
1. d_lru
2. d_sb
3. d_inode
4. d_op

Question 218
.... field in the page descriptor encodes the node and zone number to which the page frame belongs.

Question 219
.... field in dentry object provides pointer to Inode associated with filename.
Select one:
1. d_inode
2. d_sb
3. d_op
4. d_lru

Question 220
.... operation associated with file object Checks whether there is activity on a file and goes to sleep until something happens on it.
Select one:
1. ioctl(inode, file, cmd, arg)
2. poll(file, poll_table)
3. fsync(file, dentry, flag)
4. mmap(file, vma)

Question 221
SAY TRUE OR FALSE
"If the page has the Read or Execute access right but does not have either the Write or the Share access right, the Read/Write bit is set."

Question 222
.... state of dentry object indicates that the dentry object is currently used by the kernel. The d_count usage counter is positive, and the d_inode field points to the associated inode object. The dentry object contains valid information and cannot be discarded.

Question 223
Given a main memory of 4GB, determine the maximum number of page frames in this memory considering Linux default page frame size.
Select one:
1. 1Giga page frames
2. 4 Mega page frames
3. 1Mega page frames
4. 1Kilo page frames

Question 224
.... operation is used as second argument of flock system call to remove lock on the file.
Select one:
1. LOCK_SH
2. SET_LCK
3. LOCK_EX
4. LOCK_UN

Question 225
New Slab objects may be allocated by invoking the .... function.
Select one:
1. kmem_cache_free( )
2. vmalloc( )
3. vfree( )
4. kmem_cache_alloc( )

Question 226
…………………… can be used inside interrupt handlers and deferrable functions, because requesting such a mapping never blocks the current process and it uses a set of 13 windows or Page Table entries for mapping per CPU.
Select one:
1. Temporary kernel mappings
2. Contiguous Memory Allocation
3. Permanent kernel mappings
4. Non-Contiguous Memory Allocation

Question 227
"Mandatory locks are enforced even between noncooperative processes."
SAY TRUE OR FALSE.

Question 228
Categorize the ReiserFS file system into ………………………… class.
Select one:
1. Disk-based file system
2. Network file system
3. Special file system
4. None of these classes

Question 229
The ………………… function decreases the mm_users field of a memory descriptor and invokes mmdrop() function to release the memory descriptor.
Select one:
1. mmdrop( )
2. mmput( )
3. active_mm
4. mm

Question 230
The ………………………. function, which is the Page Fault interrupt service routine for the 80x86 architecture which compares the linear address that caused the Page Fault against the memory regions of the current process to check access right violations.
Select one:
1. do_munmap( )
2. vma_unlink( ) function
3. do_page_fault( )
4. do_mmap( ) function

Question 231
……………… field in the zone descriptor provides pointer to first page descriptor of the zone.
Select one:
1. zone_start_pfn
2. zone_mem_map
3. zone_active_list
4. zone_pgdat

Question 232
.... operation is used as second argument of flock system call to apply read lock on the file.
Select one:
1. LOCK_EX
2. SET_LCK
3. LOCK_UN
4. LOCK_SH
Question 233
.... operation associated with inode object creates a new hard link that refers to the file specified by old_dentry in the directory dir; the new hard link has the name specified by new_dentry.
Select one:
1. lookup(dir, dentry, nameidata)
2. follow_link(inode, nameidata)
3. link(old_dentry, dir, new_dentry)
4. mknod(dir, dentry, mode, rdev)
Question 234
Two signals which cannot be caught by a process are ....
Select one:
1. SIGSTOP & SIGKILL
2. SIGQUIT & SIGINT
3. SIGKILL & SIGCHLD
4. SIGSTOP & SIGQUIT
Question 235
Ctrl & BACKSLASH key combinations generate .... signal.
Select one:
1. SIGSTP
2. SIGKILL
3. SIGINT
4. SIGQUIT
Question 236
SAY TRUE OR FALSE
"The problem of non-allocation of high-memory page frames does not exist on 64-bit hardware platforms, because the available linear address space is much larger than the amount of RAM that can be installed in short, the ZONE_HIGHMEM zone of these architectures is always empty."
Question 237
The purpose of the .... function is to split a memory region that intersects a linear address interval into two smaller regions, one outside of the interval and the other inside.
Select one:
1. do_mmap( )
2. vma_link( )
3. vma_unlink( )
4. split_vma( )

****